Fighting Fire with Fire(fighting Foam): The Long Run Effects of PFAS Use at U.S. Military Installations

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Abstract

Tens of millions of people in the U.S. may be exposed to drinking water contaminated with perand poly-fluoroalkyl chemicals (PFAS). We provide the first estimates of long-run economic costs from a major, early PFAS source: fire-fighting foam. We combine the timing of its adoption with variation in the presence of fire training areas at U.S. military installations in the 1970s to estimate exposure effects for millions of individuals using natality records and restricted administrative data. We document diminished birthweights, college attendance, and earnings, illustrating a pollution externality from military training and unregulated chemicals as a determinant of economic opportunity.

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1 Introduction

Synthetic chemicals appear in agricultural, industrial, and household products, from pesticides and plastics to cosmetics and textiles, but their health effects are under-studied (Gold and Wagner, 2020). One such group, per- and poly-fluoroalkyl substances (PFAS), were first developed in the 1940s and eventually widely adopted. This class of chemicals is pervasive, persistent, and likely toxic—and the characteristics of their development and adoption were not unique: over the last century, tens of thousands of new chemicals entered use in the U.S. without being assessed for health and safety risks (Richter et al., 2021).

Access to clean drinking water can prevent a range of adverse health outcomes (Currie et al., 2013; Keiser et al., 2023). Ongoing monitoring efforts suggest exposure to PFAS in drinking water is widespread. The EPA's most recent wave of data collection found the chemicals[1] in 16 percent of sampled U.S. public water systems (US EPA, 2024a), and the USGS estimates tens of millions of Americans may be exposed to PFAS through reliance on contaminated groundwater (Tokranov et al., 2024). This pollution is, in part, the legacy of decades of use at U.S. military installations. Testing done in 2016 at England Air Force Base, which has not been active since 1992, found PFAS groundwater concentrations of 20,700,000 parts per trillion (ppt); the EPA's health advisory level is 70 ppt.[2] PFAS are odorless, colorless, and tasteless in water.

This study estimates the short- and long-run impacts of in utero exposure to a major, early source of PFAS pollution. We first build a timeline of the development and adoption of PFAS-based fire-fighting foams (aqueous film-forming foams, or AFFF) for military training. The new chemicals had undeniable benefits: faster, more reliable performance extinguishing fuel fires in emergencies (such as aircraft crash landings) can save lives and prevent the loss of costly equipment.[3] However, routine, intensive fire training without effluent controls led to PFAS groundwater contamination that persists to this day (Anderson et al., 2016; Ruyle et

[1] Specifically, perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS), the most widely studied compounds within the PFAS class; PFOS-based AFFF is associated with contamination from fire training activities (US EPA, 2017).

[2] EPA proposed this drinking water health advisory level for PFOA and PFOS in 2016. It is not a regulatory standard, but provides information about lifetime exposure levels above which adverse health effects can occur. In 2022, interim health advisory levels were updated to 0.004 and 0.02 ppt (PFOA, PFOS) (US EPA, 2024b).

[3] The $53 million cost of an F-15A, which entered service in 1975, or the $28 million cost of an F-16A, which entered service in 1979 (U.S. Air Force, 2023 USD), exceeds the current $11 million value of statistical life (U.S. EPA, 2023 USD).

al., 2023). We combine the timing of AFFF adoption with variation in fire training locations to estimate the effects of exposure on individual birthweights from county-level natality data and adult earnings from linked Census and Internal Revenue Service (IRS) administrative records across cohorts born in the 1970s and 1980s. We believe AFFF adoption is informative about PFAS exposure because the link from AFFF use to groundwater PFAS contamination is well established (Hu et al., 2016; Adamson et al., 2022), and much of the U.S. relies on groundwater for drinking water (USGS, 1985). Studying this early appearance of PFAS has empirical advantages: during our study period, the class of chemicals was decades away from salient regulatory and public concern, and not yet widespread in the environment and human bodies.[4]

We find birthweights fell in counties exposed to military fire training areas after the adoption of AFFF, relative to counties with installations but no fire training. Statistically significant effects lag the adoption date by several years, consistent with a hydrological delay as contamination traveled through soil to groundwater. Children born in the second half of our post-adoption period in exposed counties weighed 7.6 grams less than children in control cohorts. The magnitude of this effect is comparable to studies of in utero exposure to maternal stress (Camacho, 2008), and somewhat smaller than the 30 gram birthweight increase estimated for water treatment for lead (Dave and Yang, 2022).[5]

The harmful effect of exposure to the adoption of AFFF at military installations persists into adulthood: we find individuals born in Census tracts within 30 kilometers of sites with fire training areas had 1.7 percent lower earnings decades later and were 0.9 percent less likely to complete college. The magnitude of this long-run earnings effect is comparable to that of the original Clean Air Act (Isen et al., 2017). We incorporate multiple proxies for differences in exposure risk across sites—which depends on site characteristics, local hydrology, and local drinking water sources—and show estimated effects are generally driven by sites with more complete exposure pathways, located in more groundwater-reliant counties.

This study provides the first causal, national-scale estimates of exposure to a source of PFAS contamination on health at birth and adult earnings, and contributes to a broader

[4] In 2019, the CDC estimated 97 percent of Americans had PFAS in their blood (CDC-ATSDR, 2022).

[5] Relative to our county-level exposure measure, individuals were directly matched to their water source, reducing attenuation bias from measurement error.

understanding of the short- and long-run effects of water pollution and toxic chemicals. Correlational medical studies find mixed effects of PFAS ranging from diminished health at birth to impacts on cardiac, immune, respiratory, and other systems; these are summarized in Appendix C.

We also contribute the first economics study to estimate long-run effects of local pollution generated by Department of Defense (DOD) operations in the U.S. The military is a major domestic polluter: activities such as munitions training and propellant manufacturing generate millions of pounds of toxic releases to the environment every year (DOD, 2008). Other studies of military-related pollution include Bobonis et al. (2020), which finds a sudden decrease in Navy bombing activities in 2000 in Vieques, PR, decreased congenital anomalies, and Black et al. (2019), which finds in utero nuclear radiation exposure from testing in Norway reduced IQ scores, educational attainment, and earnings. Lleras-Muney (2010) uses moves induced by Army personnel transfers to show higher ozone pollution increases children's hospitalizations.

This research expands a "fetal origins hypothesis" literature linking early exposure to pollution with adverse outcomes throughout the life course, within which criteria air pollution has been most thoroughly studied. Air pollution exposure in utero causes worse infant health, low birth weight, and increased infant mortality (Chay and Greenstone, 2003b,a; Currie and Neidell, 2005; Currie et al., 2009, 2011; Currie and Walker, 2011; Currie et al., 2014; Greenstone and Hanna, 2014; Clay et al., 2021), and these in turn have been linked to diminished later life outcomes (Almond, 2006; Black et al., 2007; Almond and Currie, 2011). Adverse effects—including lower test scores, behavioral incidents, cognitive disabilities, grade repetition, lower earnings, and diminished opportunities for future generations—have also been documented in the short- and long-run for exposure to air pollution (Sanders, 2012; Bharadwaj et al., 2017; Isen et al., 2017; Colmer et al., 2020) and toxic chemicals (Rau et al., 2015; Persico et al., 2020; Persico and Venator, 2021; Persico, 2022; Jacqz, 2022).

The effects of water pollution are considerably less well understood, particularly at national scale and in the long run. Lead exposure is a notable exception (Clay et al., 2014; Grossman and Slusky, 2019; Dave and Yang, 2022; Fletcher and Noghanibehambari, 2023): for example, Ferrie et al. (2012) connects exposure to lead pipes to diminished cognitive scores among young men. Contaminated water, and especially chemical pollution, has been linked to short-run effects

including worse health at birth, school absences, and poor academic performance (Currie et al., 2013; Hill and DiSalvo, 2023; Marcus, 2020, 2023). Meanwhile, water purification and filtration lead to health improvements: water purification technology to reduce early-life exposure to typhoid fever, increases later-life earnings and educational attainment (Beach et al., 2016), and granular activated charcoal filtration in a community near Minneapolis decreased instances of low-birthweight and premature births (Waterfield et al., 2020).[6] Safe Drinking Water Act investments reduced concentrations of regulated contaminants, which in turn reduced adult mortality within the following decade Keiser et al. (2023).

Our findings also highlight the need for an improved understanding of risks from chemical innovation. The chemical industry boomed in the mid-twentieth century, with the production of synthetic organic chemicals rising from five to 350 billion pounds from 1935 to 1978 (Schroeder, 1983). Concerns about new chemicals led to regulation—the Toxic Substances Control Act (TSCA) was passed in 1976—but only ten percent of chemicals introduced between 1979 and 2016 faced testing (Gold and Wagner, 2020) and only five chemicals were ever restricted under TSCA;[7] the PFAS class went unrestricted for decades. Regulation affects both chemical innovation and consumer safety: stringent testing requirements can improve consumer surplus (Grennan and Town, 2020), and better design of regulations can overcome information asymmetries between manufacturers and regulators (Coria, 2018); on the other hand, regulation can spur chemical innovation (Hultgren, 2023). This study illustrates a setting with high external costs of chemical innovation.

Finally, PFAS face increasing scrutiny by environmental agencies. In March 2023, the EPA proposed a binding Maximum Contaminant Level (MCL) for PFAS in drinking water of four ppt—a limit so stringent that any detectable level of PFAS in drinking water would exceed the standard (US EPA, 2024b). Cost-benefit analysis for the rule included possible health effects in childhood and adulthood; we contribute by showing substantial earnings losses for individuals likely exposed to AFFF. In the absence of a federal MCL, some states have implemented standards, which vary in stringency.[8] As with other pollution, costly private investments can reduce exposure—but leaving protection to households risks leaving the most disadvantaged

[6] The community water had high PFAS, which filtration would have removed along with many other contaminants.
[7] Other federal programs regulate specific groups of chemicals, including food, drugs, cosmetics, and pesticides.
[8] As of December 2023, 14 states impose stricter limits than 70ppt, and 6 states mandate EPA's advisory level (ITRC, n.d.).

families behind.

2 Background: AFFF at military installations

> *Fuel fires are one of the greatest threats faced by the Navy and Air Force in their day-to-day and combat roles. ... The military's fire fighting capability (manpower and equipment) must be at a continuous state of readiness to meet these threats. Training in fire fighting skills is an essential element of the military capabilities to control fires aboard ship, on shore, and during crash/rescue operations.* (Bass, 1982)

Per- and poly-fluoroalkyl substances (PFAS) are a class of chemicals invented in the 1940s, and eventually widely adopted for use in industrial and consumer products for their heat, water, and stain resistance; non-stick cookware and water-repellent clothing are familiar applications. Their chemical stability, which makes them effective, also makes them so resistant to biodegradation that they have been nicknamed "forever chemicals." Within this large class—the EPA lists over 14,000 in a database—fluorinated surfactants[9] are a major subgroup, and the active ingredient in aqueous film forming foam (AFFF), which is used to control liquid fuel fires.[10]

AFFF initially met minimal regulatory scrutiny. The 1976 TSCA required manufacturers to notify EPA 90 days in advance of new commercial chemical production and provide internal toxicity research.[11] The burden of proving additional testing was warranted fell on the regulator, and this rarely occurred (Bass, 1982; Richter et al., 2021). To our knowledge no AFFF formulation was prohibited or restricted by the EPA in the 1970s or 1980s.

The military, which handles flammable liquids (like jet fuel) that cannot be extinguished with water alone, was the first major consumer of AFFF; a 2004 report estimates the DOD purchased 75 percent of the U.S. market (Darwin, 2011). Fire-fighting foams are particularly important for the Air Force and Navy, relative to the Army, with their greater use of aircraft and corresponding need to train for fire suppression when planes crash.

[9] A surfactant reduces the surface tension of a liquid in which it is dissolved.

[10] A blanket of sprayed foam provides fire suppression by preventing oxygen from reaching burning fuels. Early formulations of AFFF were called "light water," because the surfactant-water mixture was observed to float on the surface of gasoline (MacKay, 1991).

[11] For example, in 1981 3M submitted a TSCA Section 8(e) Risk Notice for several fluorochemicals, including AFFF formulations (FC-203, FC-203A, FC-206A, FC-600, and FC-780B), reporting toxic effects in rats and mentioning fluorochemicals detected in employees' blood, but noting no observed health issues or disease patterns linked to exposure.

2.1 Invention and adoption

AFFFs were among the earliest commercially viable uses of PFAS (MacKay, 1991). Developed through a partnership between 3M and the U.S. Navy's Naval Research Laboratory,[12] they quickly proved to be more heat resistant, effective at lower concentrations, and longer-lasting for burnback prevention than their predecessors (Geyer et al., 1979; Peterson et al., 1967).[13]

The military adopted AFFFs after years of development. Each AFFF formula had to meet a performance standard (Military Specification, or MILSPEC) to be qualified for use at DOD installations. Frequent updates to the early MILSPEC drove new product offerings.[14] The exact PFAS formulations in early MILSPEC-compliant AFFFs were proprietary trade secrets and remain mostly unknown today (Place and Field, 2012).[15]

"The 1970s were boom years for ... AFFF products", according to a history of 3M in the 20th century (MacKay, 1991). October 1971 is the earliest patent date for a PFAS compound commonly detected in groundwater at military installations in Ruyle et al. (2021). The 1971 DOD appropriations hearings discussed the costs of switching to AFFF.[16] A 1972 report states "AFFF has now widely replaced protein-type foam in the Navy" (Gipe and Peterson, 1972),[17] and a 1973 report claims "aqueous film-forming foams have or are currently replacing the protein-type foams as the primary fire-fighting agent at most Air Force installations" (Kroop and Martin, 1974). The Army likely adopted AFFF more slowly, focusing on dry chemicals for ammunition fires.[18] Civilian adoption followed later and has been less associated with contamination (Federal Aviation Administration, 1985; National Academies of Sciences,

[12] The deadly 1967 fire on the USS Forrestal spurred investments in fire fighting training and technology.

[13] Until the introduction of AFFFs, the DOD relied on animal protein-based fire-fighting foams. These were more biodegradable than AFFF (Geyer, 1973), but likely shared other components, as early AFFFs incorporated PFAS into existing formulations (Kempisty et al., 2018). In addition to surfactants, AFFF concentrate contains water, solvents, and other additives. Modern PFAS-free foams are less toxic (Kempisty et al., 2018; Ateia et al., 2023).

[14] In 1969, 3M introduced FC-195, the first seawater-compatible AFFF, under a Navy MILSPEC, followed by FC-196, FC-199, and FC-200. These were made obsolete in 1972 with a much less corrosive but highly toxic alkaline AFFF formula (Gipe and Peterson, 1972; Salazar and CA, 1985; Kempisty et al., 2018; Dlugogorski and Schaefer, 2021).

[15] 3M's early military-inventory AFFFs were PFOS-based, while other manufacturers' (including Ansul and National Foam) contained precursors of PFOA and PFOS (Buck et al., 2011).

[16] A FY 1971 Air Force justification titled *Conversion to New Type Foam for Aircraft Fires* is the earliest indication we found of a large purchase of AFFF: "This new requirement is for aqueous film forming foam which replaces the old protein type foam for fighting aircraft fires. It is 30 percent more effective and also more expensive ... The annual requirement is 1.0 million versus 1.5 million gallons for the protein type." (p. 127).

[17] It was initially prioritized for shipboard fire suppression systems (Gipe and Peterson, 1972), and later adopted for training activities.

[18] Most Army and National Guard soldiers trained with AFFF at Air Force bases. In the 1980s, the Army tested AFFFs on ammunition fires (Finnerty, 1982).

Engineering, and Medicine, 2023).

After AFFF was adopted, it was incorporated into aircraft crash fire training drills. Designated Fire Training Areas (FTAs) saw particularly heavy use of fire-fighting foams—both before and after the adoption of AFFF. A typical drill involved filling open, unlined pits with flammable liquids and lighting these on fire, which trainees practiced extinguishing. Effluent, including AFFF, was washed into holding ponds or the surrounding environment to reset the burn pit (Anderson et al., 2016).[19] Systematic records of training intensity are scarce, but high-use FTAs saw weekly drills using hundreds of gallons of fuel and dozens of gallons of AFFF concentrate (Taylor, 1976).

2.2 Contamination

Military AFFF use between 1970 and 1990 was a major source of historic PFAS contamination in groundwater (Houtz et al., 2013; Anderson et al., 2016; Moody and Field, 2000).[20] PFAS contaminated groundwater both up- and downstream of military installations through runoff, drainage, and leaching into soil (Filipovic et al., 2015; Ahrens et al., 2015; Hu et al., 2016; Charbonnet et al., 2021); site-specific contamination levels depend on release quantity, AFFF formulation, and local hydrology and meteorology (Awad et al., 2011; Quezada Davalos et al., 2022).[21] Studies suggest PFAS take years to move through soil to groundwater, where they accumulate and persist (Bove et al., 2020; Ruyle et al., 2023; Meeks et al., 2023).

In 2018, the DOD acknowledged historic FTAs were of primary concern for PFAS contamination (Copp, 2018). Groundwater at Air Force and Navy installations with FTAs often exceeds the EPA's 70 ppt advisory level by many orders of magnitude (Figure A.1). By 2023, the DOD was distributing "alternative water" to communities surrounding 53 installations with PFOA and PFOS levels exceeding 70 ppt and committed to following forthcoming EPA drinking water standards (US DOD, 2023).

[19] Discharging FTA effluent to sewage systems was discussed in the mid-1970s, but this was infeasible at many Navy installations ("smaller operations do not have the economic resources to provide, operate, and maintain sophisticated treatment technologies", Bass (1982)), and an Air Force report suggests effluent controls were not widely adopted (US DOD, 1991). Most FTAs likely remained unlined during our study period.

[20] Appendix B.2 discusses other historic PFAS sources.

[21] Early AFFFs used long-chain PFAS, which are less mobile in the environment and accumulate in soil and groundwater (Hu et al., 2016; Reinikainen et al., 2022). Wetter climates dilute and drier climates concentrate environmental PFAS (Adamson et al., 2022). AFFF releases are not linked to windborne transport, unlike sites that manufactured or disposed of PFAS (Shin et al., 2011; D'Ambro et al., 2021).

PFAS bioaccumulates in humans, suggesting children who are in utero to mothers who have been drinking contaminated water for longer may be more at risk for adverse health outcomes.[22] In 2019, the CDC found levels of PFAS in blood four–five times above the national average among individuals in communities surrounding eight Air Force bases (CDC-ATSDR, 2022).

3 Data

To characterize and estimate the effects of exposure to the adoption of AFFF, we combine DOD installation locations with natality records and individual administrative data. Data sources are discussed in greater detail in Appendix B.

3.1 Military installations

The DOD released a list of candidate military installations suspected for PFAS contamination in July 2021 (US DOD, 2021b), and we combine this with an earlier DOD report that inventoried former, active, and new FTAs at each site (US DOD, 2014). We match these to installation boundary shapefiles, and then geolocate installations, identifying counties containing installations, and Census tracts overlapping buffers of increasing distance (up to 30 kilometers) to installations.

We use the presence of an FTA at an Air Force or Navy installation as a proxy for likely PFAS contamination from AFFF in the 1970s and 1980s.[23] There is also no systematic data on installations' AFFF adoption dates nor point locations of FTAs.[24] Fifty-three percent of the installations in our study never had an FTA. Figure A.2 shows the location of counties containing military sites with and without FTAs.

In 1975, a DOD-wide initiative called the Installation Restoration Program (IRP) was created to investigate hazardous pollution at military installations. We collect two pieces of information from the IRP reports: corroboration of the presence of a FTA, and mentions of training as an indicator that the FTAs were regularly used.

In the 2010s, the DOD began testing groundwater for PFAS and releasing site reports; both

[22] The half-life of PFAS is three–five years in the body (Olsen et al., 2007).

[23] We know of no historic measurements of PFAS in groundwater at or around DOD installations before a 2003 site study (Moody et al., 2003).

[24] Early on, the decision to adopt and/or train with AFFF was left to installation commanders (MacKay, 1991).

ongoing efforts prioritize sites with expected contamination. These reports detail contamination concerns, off-site PFAS migration pathways, and affected populations. We standardize exposure risk using a rubric; low scores can reflect low risk or missing information. Though reports incorporate PFAS contamination occurring after our study period, the reports may be informative about historic variation in contamination.

We match installations to closure dates (an outcome of the congressional process called Base Realignment and Closure (BRAC), which aimed to reduce operating costs), and Superfund listing dates. Both outcomes are rare over the study period. We use the U.S. Geological Survey (USGS) National Hydrography Dataset to identify the direction of likely surface flow from military installations, and define an FTA-exposed downstream tract as one intersected by a downstream flowpath from an installation with an FTA.

Summary statistics for the sites in our dataset are presented in Panel D of Table A.1.

3.2 Natality, administrative, and county data

Observations of births come from the Natality Birth Data from the National Vital Statistics System. These are individual birthweights, demographic data, and county for each year in our study period, 1969–1988. Summary statistics for births in our sample are presented in Panel A of Table A.1.

Administrative data from the Census Bureau and IRS contain every individual born from 1969–1989 who completed the American Community Survey (ACS) in 2005–2019 and can be linked to parents who filed taxes in 1974, 1979, or 1984. We identify individuals born near installations during the study period using parents' tax returns, which are geocoded at the home address level and aggregated to Census tract boundaries, and contain dependent children's SSNs. Because the individuals in our sample had a documented, tax-filing parent, we expect them to be economically advantaged relative to the general population. We expect these parents to include both military personnel and members of surrounding communities.

Linked individual responses to the ACS in any year between 2005 and 2019 provide demographic characteristics and educational attainment. IRS records for the 2015–2019 tax years, when individuals born during our study period are 26–50 years old, provide Adjusted Gross Income (AGI) from 1040s and total earnings from W-2s. Summary statistics for individuals

in our administrative data are presented in Panel B of Table A.1.

Time-varying county economic indicators are from the Bureau of Economic Analysis. USGS provides county population shares in 1985 served by groundwater (public-sourced or private domestic wells) and surface water. County-level precipitation and temperature data are from Schlenker and Roberts (2009), and wind data are from the North American Regional Reanalysis data set for 1979–1990. Total suspended particluates (TSP) and county-level Clean Air Act nonattainment status are from the EPA, and Clean Water Act grants are from Keiser et al. (2023). Summary statistics for counties in our sample are presented in Panel C of Table A.1.

4 Specification and results

We pair the quasi-exogenous timing of the adoption of AFFF by the Navy and Air Force with variation in the location of FTAs to estimate the dynamic effect of exposure to PFAS-based AFFF. Individuals born near installations without FTAs provide a counterfactual for short- and long-run outcomes.

We restrict our study to individuals born in places near any military installation, who may be different from the general population in unobservable ways. Within this group, a causal interpretation of our estimates requires FTA locations and AFFF adoption timing are uncorrelated with local trends in all other determinants of birthweight and earnings.

The DOD's adoption of AFFF occurred rapidly in the early 1970s. It is unlikely the timing of AFFF adoption was driven by trends in the health or sociodemographics of populations living near military sites, nor was it anticipated or salient to households: fire training with other chemicals began before the 1970s, and public awareness of the health effects of PFAS came decades later.

We treat all installations with FTAs as adopting simultaneously in 1972. Changes over time in AFFF availability and formulation affected all installations. There is little evidence routine use began earlier, and a lack of site-specific records prevents assigning later adoption dates. Delayed adoption would bias estimate magnitues downward for older cohorts. We do not expect immediate effects of AFFF adoption because PFAS contamination and accumulation in water take years.

Individual characteristics are very similar across FTA exposure status, but FTA-exposed

counties had somewhat higher populations and per capita incomes from 1969–1972 (Table A.1). An analysis of Air Force FTAs concluded their construction dates were not geography- or mission-related (Brewer, 1988), and a 1991 report on FTA site selection criteria does not mention region, mission, or installation characteristics (US DOD, 1991).

Empirical specifications estimate the dynamic effect of the adoption of AFFF at military installations on individuals born nearby. The small number of cohorts observed before the 1972 reference year will limit our ability to visually assess pre-trends, as in Isen et al. (2017). Because our natality and earnings datasets differ in their geographic precision, we use different specifications for each, and discuss below how we reconcile estimates. Our panel is balanced and treatment timing is not staggered, so we avoid the issues that motivate recent advances in the difference-in-differences literature (Roth et al., 2023), and implement "canonical" two-way fixed effects specifications.

4.1 Birthweight effects

We estimate birthweight effects with

$$Y_{ift} = \sum_{t=1969}^{1988} \tau_t \, AnyCountyFTA_{if} + \mathbf{X}'_{ift}\gamma + \mu_f + \theta_{srt} + \epsilon_{ift}, \tag{1}$$

where Y_{ift} is the birthweight of child i born in county f in year t and $AnyCountyFTA$ is equal to one if the county contains one or more military installations with an Air Force or Navy FTA. The estimating sample consists of children born from 1969–1988 in counties with at least one installation. X_{ift} contains the mother's age, annual county average temperature, average precipitation, logged county population, income per capita, and employment, and separate shares of installations in the county closed or Superfund-listed in t. Child sex-race-birth year fixed effects θ_{srt} capture time-varying changes affecting demographic birth cohorts in our sample, and county fixed effects μ_f capture time-invariant differences in child outcomes across counties. Conditional on these controls, the coefficients of interest τ reflect the difference in birthweights across cohorts born in counties with and without FTAs.

PFAS exposure likely increased in FTA-exposed locations over the study period. In settings where pollution falls, differences in outcomes across cohorts born just before and after the decrease reflect the effect of removing in utero exposure (as with Clean Air Act's nonattainment

counties in Isen et al. (2017)). When pollution rises, the oldest cohorts do not experience pollution in utero, but all cohorts may be affected by contaminated groundwater after birth. Our coefficients of interest reflect these cohort-specific effects of exposure to AFFF adoption relative to the 1972 birth cohort, who we expect avoided in utero, but not later-childhood, exposure.

We find the presence of an FTA begins to decrease the birthweight of children in affected counties starting in the late 1970s (Figure 1). As discussed above, a delayed effect is consistent with some years of PFAS transport to and accumulation in groundwater. We also use an otherwise-identical difference-in-differences specification with two post-period coefficients (grouping birth years 1973–80 and 1981–88) to summarize effects. Throughout, we report an average exposure effect for the latter of the two post-periods; the average birthweight effect defined in this way is –7.6 grams. In Figure A.3, we present event studies for alternative outcome variables: gestational length, a binary indicator for low birthweight, and a binary indicator for prematurity; these provide consistent evidence for adverse birth effects.

4.2 Long-run effects

The linked administrative data are more geographically precise: we observe Census tract around the time of birth, rather than county of birth. We define three treatment variables: $AnyCountyFTA$ is as above; $AnyTractFTA$ is one for tracts within 30km of any military installation with an Air Force or Navy FTA; and $InvDistFTA$ is $5/d$, where d is each tract's distance d to the nearest FTA using 5km, 10km, 15km, 20km, and 30km buffers. $InvDistFTA$ takes a maximum value of 1 when a tract's nearest FTA is within 5km and a value of 0 when a tract has no FTA within 30km.

We estimate earnings effects with

$$Y_{ict} = \sum_{t=1969}^{1989} \tau_t \, AnyTractFTA_{ic} + \mathbf{X}'_{ict}\gamma + \mu_c + \theta_{srt} + \epsilon_{ict}, \tag{2}$$

where Y_{ict} is the logged earnings, observed in 2015–2019, of an individual i born in tract c in year t. Subscripts for the tax year of reported earnings are omitted for clarity, but most individuals have multiple earnings observations (averaging 4.8 out of 5 possible years). The estimating sample depends on the treatment variable used: when the treatment is $AnyCountyFTA$, we

include all individuals born in counties containing installations, as in the birthweight specification; when the treatment variable is $AnyTractFTA$ or $InvDistFTA$, the estimating sample is all individuals born in tracts within 30km of an installation. X_{ict} is as above, as well as the interaction of year of birth with (if any) the earliest year of a nearby installation closure, and the interaction of year of birth with (if any) the earliest year of a nearby installation Superfund listing. Child sex-race-birth year-tax year fixed effects θ_{srt} capture time-varying effects on birth cohorts in our sample, and tract fixed effects μ_c capture time-invariant differences in child outcomes across place of birth (for comparability, the county-level specification follows Equation 1 and uses county FE). As above, the coefficients of interest τ reflect differences, conditional on controls, in outcomes between birth cohorts in FTA-exposed tracts relative to those proximate to military installations with no FTAs.

Earnings effects have a similar pattern to the birthweight estimates: we find lower long-run earnings of cohorts born within 30km of an installation with an FTA, with a delayed average effect, as defined above, of –1.7 percent. The county-level treatment variable provides consistent but noisier estimates of this effect. The inverse-distance treatment variable suggests effects are strongest for individuals born in tracts nearest to FTA-exposed installations. ACS-reported estimates of work hours suggest earnings effects are driven by lower hourly wages rather than fewer hours spent working.

The literature estimating the effect of birthweight on earnings helps reconcile our short- and long-run effect magnitudes. Black et al. (2007) finds a 10 percent increase in birthweight causes a 1 percent increase in earnings, while Nakamuro et al. (2013) finds no effect on earnings. We find an economically insignificant effect on birthweight (less than 1 percent of a typical birthweight), but a meaningful effect on earnings (–1.8 percent in the more comparable county specification), orders of magnitude larger than expected following Black et al. (2007). This suggests AFFF exposure may affect children's health and cognitive development without diminishing birthweight, consistent with Persico et al. (2020), which finds larger cognitive and behavioral outcomes from in utero exposure to toxic waste sites than implied by corresponding birthweight effects.

There are several plausible mechanisms for the persistent effect of PFAS exposure on earnings. Early life exposure to toxic substances has been linked to lasting adverse cognitive

effects, and medical studies of PFAS suggest diminished executive functioning. We replace logged earnings in equation 2 with a binary variable indicating whether the individual had completed college at the time of the ACS. We find evidence consistent with the possibility that PFAS exposure diminishes human capital formation: later-treated cohorts (as above) are 0.9 percent less likely to have completed college (Figure 2E). Estimates of the college wage premium suggest this channel can explain a meaningful part of our earnings effect (Deming, 2023; Ashworth et al., 2021). We find no comparable statistically significant effects for high school completion rates.

Other mechanisms are more challenging to explore given available data, and warrant further study. In particular, PFAS exposure has been associated with an increased risk of health effects, including several types of cancer and cardiovascular diseases that may manifest in adulthood and limit opportunities in the labor market.

4.3 Exposure risk

We expect differences in FTA use and characteristics to influence PFAS contamination in local groundwater, and differences in groundwater consumption to influence PFAS exposure. We use various data sources to approximate differences in exposure risk, though each has limitations, and most create selected samples. Together they corroborate our expectation that higher contamination likelihoods are associated with larger treatment effects.[25] We maintain our treatment variable throughout (except in the final two specifications, described below), re-estimating equations 1 and 2 with sample restrictions. We describe findings below, and present event studies and DID coefficients in Figures A.4 and A.5 and Table A.2.

The DOD's PFAS-focused site investigations include information about drinking water risk, and recent groundwater measurements may be a proxy for historic contamination. We find post-1980 cohort DID coefficients are only statistically significant for individuals near sites with above-median risk scores[26] (–9.97 grams and –1.74 percent), but no conclusive evidence current PFAS groundwater measurements are similarly associated with the strength of estimated effects on birthweights or earnings.

[25] This descriptive heterogeneity does not imply causality, as risk factors are correlated with each other and possibly other unobserved characteristics that mediate PFAS exposure effects.

[26] To construct risk scores, we apply a standardizewd rubric to the DOD's site investigation reports; see Appendix B for details.

Consistent with a PFAS exposure pathway through groundwater rather than surface water, estimated effects on birthweights and earnings are driven by individuals born in counties with above-median reliance on groundwater (public-sourced or private wells). The post-1980 cohort DID coefficients are only statistically significant for this half of the estimating sample (–8.03 grams and –2.42 percent).

We find estimates are generally unchanged when we drop counties or tracts with FTA status in our main specification but no IRP record of an FTA; and we find estimates are also comparable for a further restriction, dropping counties or tracts without an IRP report that mentions a training frequency at the FTA. We also do not find sites with higher training frequencies, as described in an IRP report, drive our estimates.

We find no evidence that downstream tracts are driving our effects, possibly because our definition of "downstream" tracts is imprecise.[27] We similarly find no evidence downwind tracts are more affected by FTAs.

Taken together, variation across sites in PFAS contamination and exposure risk provides either corroborating (as-expected larger estimated effects for higher proxies for risk) or inconclusive evidence that AFFF adoption harmed short- and long-run outcomes through the channel of local groundwater contamination.

4.4 Interpretation and threats to identification

Changes in other sources of pollution correlated with AFFF-exposed military sites (or counties) during our study period would bias our estimates, either through direct effects or sorting that altered sociodemographic composition of mothers. Of greatest concern would be upward bias in estimate magnitudes from increasing exposures at FTA-exposed locations.

Our results are robust to the exclusion of counties in which PFAS manufacturing sites (the other major source of historic pollution) were located (Figure A.7). There is limited information on when PFAS was incorporated into consumer products, but it is likely exposure from other sources increased with birth year for our entire study population (exposed and control). This is not a threat to identification, but suggests careful interpretation: no population in the

[27] We also found the average initial distance from a sampled point to an intersection with USGS's network of surface streams is over 2km—suggesting soil leaching and not transport via surface streams and rivers as a likely pathway for contamination; the direction of PFAS transport in soil and groundwater also may not correspond to the surface flow gradient.

United States over the past half-century represents a true zero-exposure control (Blake and Fenton, 2020).

Changes in The Clean Air and Water Acts (CAA and CWA) brought about meaningful changes in exposure to other sources of pollution in a county during our study period, with associated public health benefits (Chay and Greenstone, 2003a; Isen et al., 2017; Keiser et al., 2023). We find FTA-exposed counties were slightly less likely to receive CWA grants, but more likely be in CAA non-attainment, with corresponding air quality improvements in the 1970s (Table A.1). Results are robust to including county annual average TSP as a control (Figure A.7), but note this drops the many installations in rural counties. (Brewer, 1988) found no time trends in air emissions from FTAs, and a 1993 report suggests training practices were gradually changing to reduce smoke over time (US EPA, 1993).

FTA-exposed sites were more likely to be Superfund-listed, but all listings occurred after 1983, and all but 27 after 1987. Results are robust to dropping Superfund-affected counties (Figure A.7). Data on other contemporaneous toxic chemical pollution are limited. Treatment and control counties were comparably represented in the Toxics Release Inventory in its first full reporting year (1988; see Appendix B).

The BRAC process may have had spillover economic effects (Dardia et al., 1996; Krizan et al., 1998; Hooker and Knetter, 2001). FTA-exposed sites were more likely to be closed, but closures came in 1988 or after our study period. Results are robust to excluding BRAC-affected sites (Figure A.7). We test sensitivity to the exclusion of each military branch. Results are robust to the separate exclusions of installations run by the Army (though this greatly reduces the size of the control group) and Navy, but not the Air Force, indicating Air Force AFFF use drives our main estimates (Figure A.7).[28]

Linked administrative data may have measurement error in assigned places of birth; this is not a threat to identification unless it varies systematically by treatment status. Our results are attenuated when we include individuals with inconsistent state of birth across records (Figure A.8). Results are robust to changes to demographic fixed effects, exclusion of controls for BRAC and Superfund dates, inclusion of state-year trends, and the use of regression weights, either by birth cohort size or number of years of observed earnings (Figure A.10), as well as the

[28] Navy FTAs may have directed effluent into harbors, preventing some groundwater contamination; there are also many more Air Force installations in our panel.

inclusion of the full set of earnings values, and noisy but qualitatively similar when earnings are in levels (Figure A.9).

Finally, we assess the parallel trends assumption more generally. The AFFF adoption date does not predict statistically significant divergences in county-level population or income across treatment and control counties, but FTA-exposed counties saw suggestively greater population growth and lower per capita income for the post-adoption years (Figure A.11). Following Rambachan and Roth (2023), we characterize the magnitude of violations of post-period parallel trends, relative to pre-period, above which the confidence interval on the average effect for the 1981–1988 cohort contains 0. The breakdown values of 0.1 suggest estimates are somewhat sensitive to violations of the parallel trends assumption (Figure A.12).

5 Discussion

Quantifying the short- and long-run consequences of in utero exposure to the adoption of AFFF at U.S. military installations advances our understanding of the economic effects of a major PFAS source, highlights the role of the military as a domestic polluter, and illustrates externalities from the unregulated development, use, and release of novel chemicals.

We find cohorts born near installations with fire training areas after the adoption of AFFF—which contaminated groundwater with PFAS—were born with lower birthweights, were less likely to complete college, and earned less as adults. A back-of-the-envelope estimate combining our delayed earnings effect ($946 per year for an affected individual) and the number of affected individuals (an average 1.17 million births per year from 1981–1988 in FTA-exposed counties) implies aggregate lost earnings of nearly $1 billion per year (2019 USD).

Decades of AFFF releases across hundreds of sites mean mitigation will be long and costly. The DOD acknowledges PFAS pollution is so widespread that fully characterizing the problem will take years, and remediation decades (US DOD, 2021a, 2024b). While PFAS contamination has become recently salient, there remains an orders-of-magnitude gap between the number of chemicals in active use in the U.S. and the number for which health risks are well understood or managed. This study can be seen as a cautionary tale—using chemicals with unknown health risks can have reverberating impacts not fully observed for decades.

6 Figures

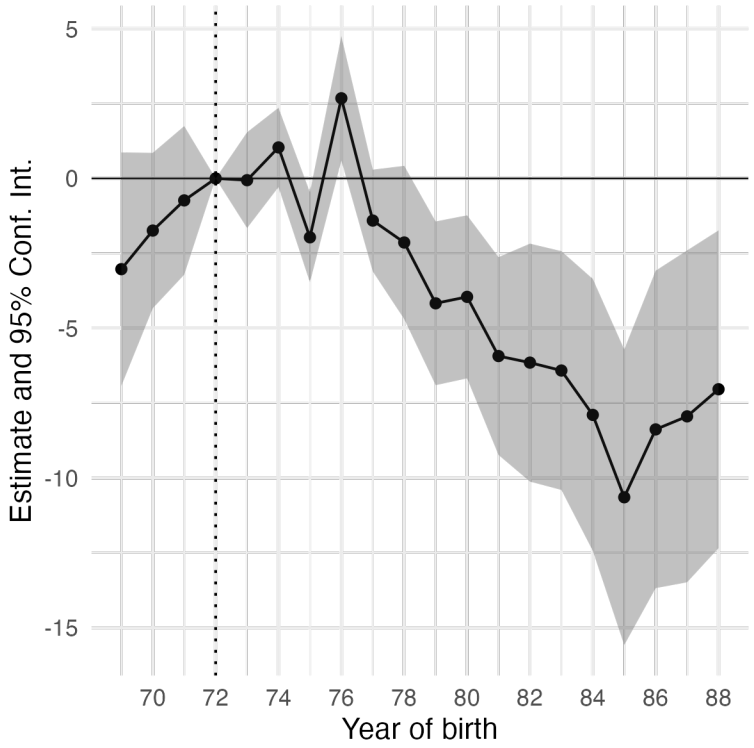


Figure 1: Birthweight effect

Note: This figure plots event study coefficients for cohorts born from 1969–1988, corresponding to Equation 1. The dependent variable is individual birthweight in grams, the treatment variable is equal to 1 in FTA-exposed counties, and the estimating sample is individuals born in counties containing any military installation. Regression controls include child and mother demographics, birth year-varying county weather and economic indicators, county fixed effects, and child sex-race-birth year fixed effects; see text for details.

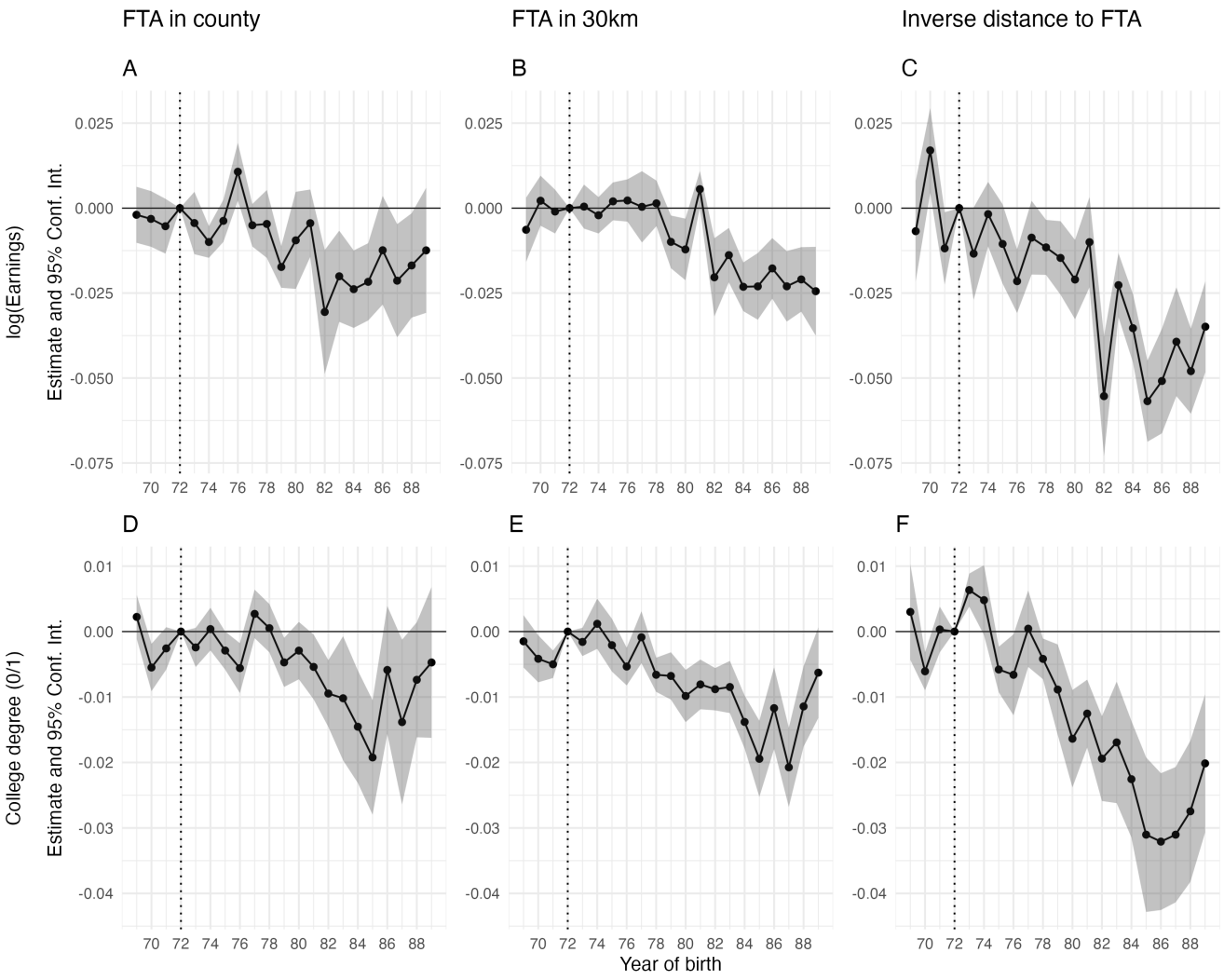


Figure 2: Earnings and college completion effects

Note: This figure plots event study coefficients for cohorts born from 1969–1989, corresponding to variants of Equation 2. The dependent variable in the first row is logged earnings, observed from 2015–2019, in 2019 USD. The dependent variable in the second row is a binary indicator for college completion, observed in the 2005–2019 ACS. In column 1, the treatment variable is equal to 1 in FTA-exposed counties and the estimating sample is individuals born in counties containing any military installation. In columns 2 and 3, and the estimating sample is individuals born within 30km of any military installation. In column 2, the treatment variable is equal to 1 in FTA-exposed tracts. In column 3, the treatment variable is equal to $5x$ the inverse distance to an FTA-exposed installation (distance can take the value of 5, 10, 15, 20, or 30km) or 0 for tracts near non-exposed installations, so the coefficients can be interpreted as the effect of being in utero in a tract within 5 km of an FTA-exposed installation. Regression controls include birth year-varying county weather and economic indicators, county (column 1) or tract (columns 2 and 3) fixed effects, and either sex-race-birth year-tax year fixed effects (row 1) or sex-race-birth year-ACS year fixed effects (row 2); see text for details.

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A Supporting figures and tables

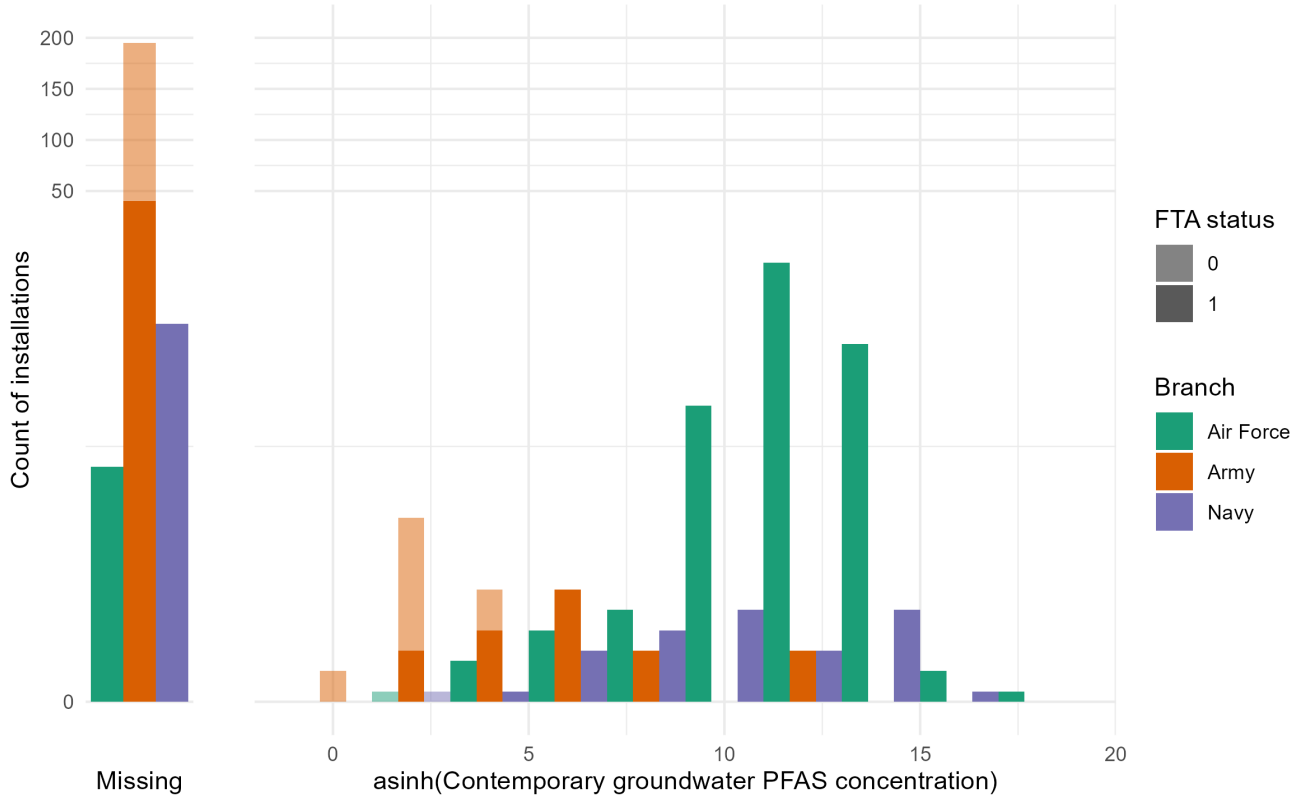


Figure A.1: Recent PFAS by installation branch and FTA status

Note: Bar heights indicate counts of U.S. military installations at binned levels of inverse hyperbolic sine-transformed measured PFAS groundwater concentrations (ppt) from the 2010s; color and shading indicate military branch and FTA presence. FTA status was coded by the authors from DOD records. Linked groundwater sampling was conducted by the DOD in the 2010—20s. At many installations, groundwater has not yet been tested for PFAS (denoted "Missing"); note compression of y-axis.

Table A.1: Balance statistics, 1969–1972

	(1) Air Force or Navy FTA	(2) All other installations	(3) Out of sample
Panel A: Births			
Birthweight (grams)	3,280 (586)	3,270 (591)	3,280 (591)
Share LBW	0.0779	0.0808	0.0789
Gestational length (weeks)	39.5 (2.8)	39.5 (2.86)	40.1 (6.24)
Share premature	0.0926	0.0949	0.0922
Mother's age	24.7 (5.49)	24.5 (5.48)	24.7 (5.66)
Share male	0.513	0.513	0.513
Share white	0.825	0.806	0.838
Births	2,271,603	1,328,773	5,144,474
Panel B: Adults			
Median earnings	55,650	54,340	49,160
Mean earnings	68,910 (59,080)	67,260 (57,680)	60,800 (52,730)
Share white	0.873	0.865	0.846
Share not Hispanic	0.902	0.941	0.935
Share with HS degree	0.782	0.787	0.773
Share with college degree	0.346	0.352	0.288
Adults	301,900	174,000	2,046,000
Panel C: County economic and environmental characteristics			
Income per capita	14,000 (2,390)	12,900 (2,720)	11,500 (2,800)
Population	350,000 (719,000)	149,000 (278,000)	67,400 (238,000)
Employment/pop.	0.456 (0.108)	0.432 (0.134)	0.406 (0.0981)
Temperature (deg. C)	13.5 (4.99)	12.2 (4.43)	12.1 (4.66)
Precipitation (cm.)	2.68 (1.2)	2.98 (1.2)	2.79 (1.06)
Share CAA non-attainment	0.67	0.602	0.528
TSP ($\mu g/m^3$)	79.9 (29.3)	84.5 (61.1)	78.6 (44.4)
Share county-yrs. w. TSP	0.539	0.404	0.179
CWA grants	1.04 (1.23)	1.12 (1.49)	1.25 (1.48)
Share served by PWS SW	0.406 (0.372)	0.336 (0.341)	0.261 (0.317)
Counties	181	259	3,107
Panel D: Site characteristics			
Share Air Force	0.675	0.114	
Share Navy	0.325	0.119	
Share Army and others	0	0.767	
Share ever Superfund	0.298	0.209	
Superfund year, among listed	1990 (4.43)	1990 (4.87)	
Share ever closed	0.298	0.201	
Closed year, among closures	2000 (6.58)	2000 (7.56)	
Share with IRP report	0.969	0.117	
Share with IRP FTA	0.851	0.354	
Share with IRP training	0.623	0.0194	
Risk score	2.72 (1.98)	1.56 (1.55)	
Share with risk score	0.899	0.672	
Current PFAS (ppt)	417,000 (1,760,000)	37,700 (137,000)	
Share with current PFAS	0.737	0.289	
Site area (sq. km.)	136 (887)	122 (608)	
Sites	228	412	

Note: Continuous variables are summarized as *mean (SD)* and binary outcomes are summarized as shares. Panel A uses natality data and Panel B summarizes administrative data, both for the 1969–1972 (pre-treatment) birth cohorts; *mean earnings* omits wages reported as 0 or above the 99th percentile by age. Panel C reports county-level economic indicators (BEA), TSP (EPA), counts of Clean Water Act (CWA) grants (Keiser et al., 2023), and population shares served by surface-sourced public systems in 1985 (USGS); time-varying data are similarly restricted to 1969–1972. Panel D reports installation characteristics.

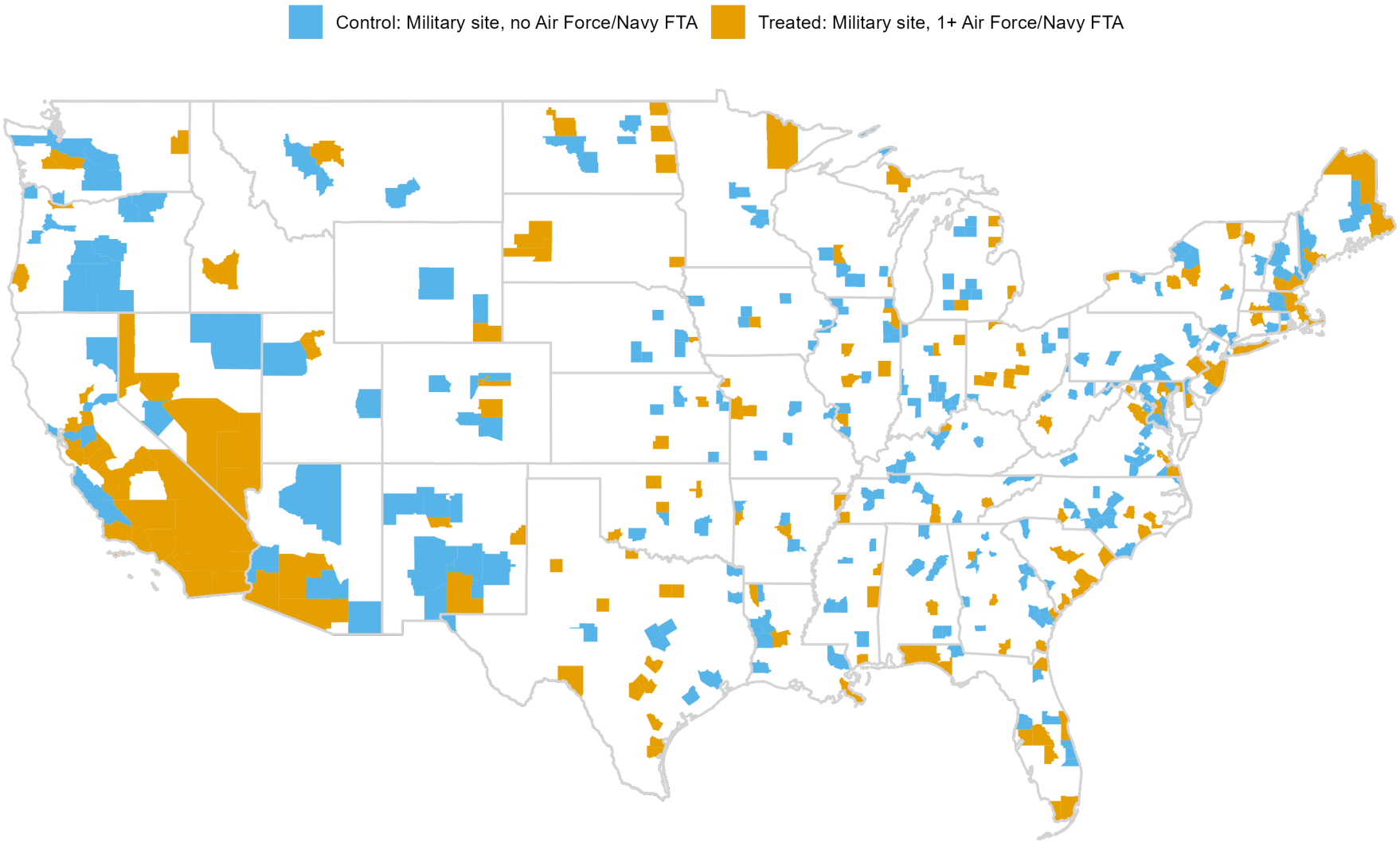


Figure A.2: Counties included in estimation

Counties identified by the DOD for possible contamination are shaded. Counties containing military sites but no FTAs are blue. Counties containing military sites with one or more FTAs are orange.

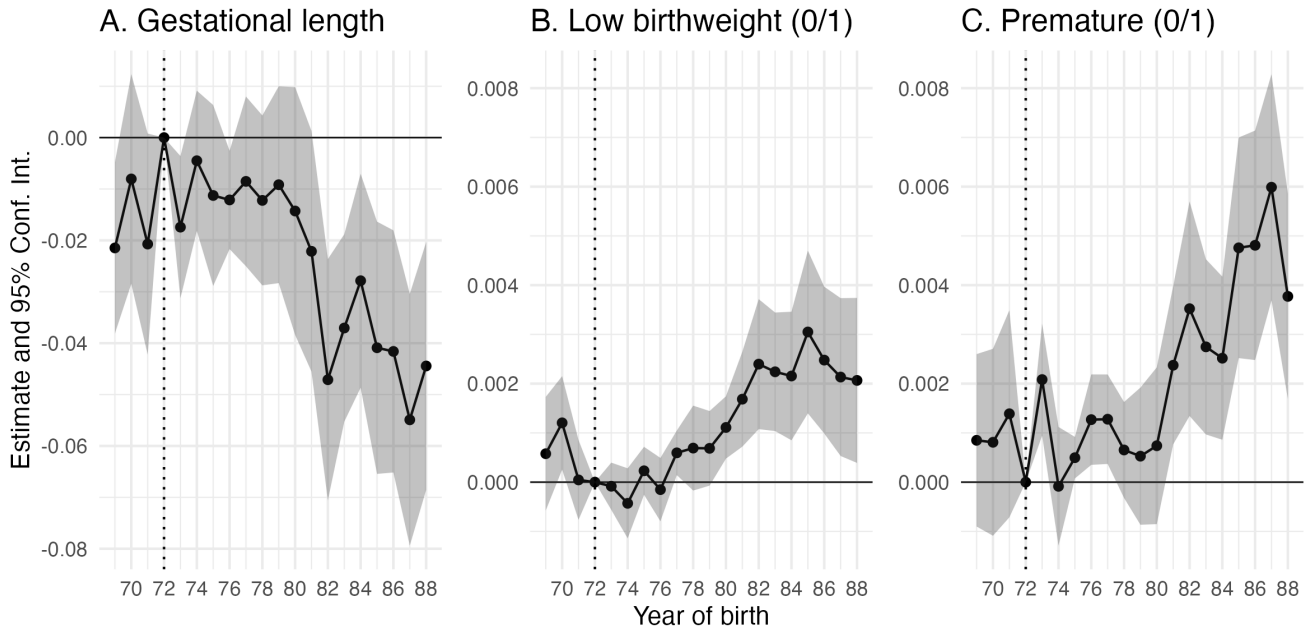


Figure A.3: Additional natality outcomes

Note: This figure plots event study coefficients for cohorts born from 1969–1988, corresponding to Equation 1. The dependent variable is gestational length in weeks in Panel A, a binary indicator for low birthweight (under 2,500 grams) in Panel B, and a binary indicator for prematurity (before 37 weeks) in Panel C. Due to missing data, outcomes related to gestational length are estimated on a somewhat smaller (and potentially selected) set of observations. The treatment variable is equal to 1 in FTA-exposed counties, and the estimating sample is individuals born in counties containing any military installation. Regression controls include child and mother demographics, birth year-varying county weather and economic indicators, county fixed effects, and child sex-race-birth year fixed effects; see text for details.

Table A.2: Exposure heterogeneity DID coefficients (post-1980 cohorts)

Sample restriction	Birthweight effect	Earnings effect
None	–6.15 (3.11)	-0.0171 (0.0050)
High risk score	–9.97 (3.87)	–0.0174 (0.0050)
Low risk score	–0.94 (3.36)	–0.0129 (0.0079)
High current PFAS	–6.57 (4.55)	–0.0224 (0.0062)
Low current PFAS	–5.49 (3.09)	–0.0160 (0.0065)
Low SW	–8.03 (4.42)	–0.0242 (0.0086)
High SW	–3.88 (4.30)	–0.0090 (0.0058)
IRP-confirmed FTA	–6.83 (3.21)	–0.0187 (0.0050)
IRP-confirmed FTA use	–3.86 (3.30)	–0.0188 (0.0053)

Note: This table presents average treatment effects (defined as the DID coefficient for cohorts born in the second half of the treatment period, *i.e.*, after 1980) corresponding to the event studies presented in Figures A.4 and A.5.

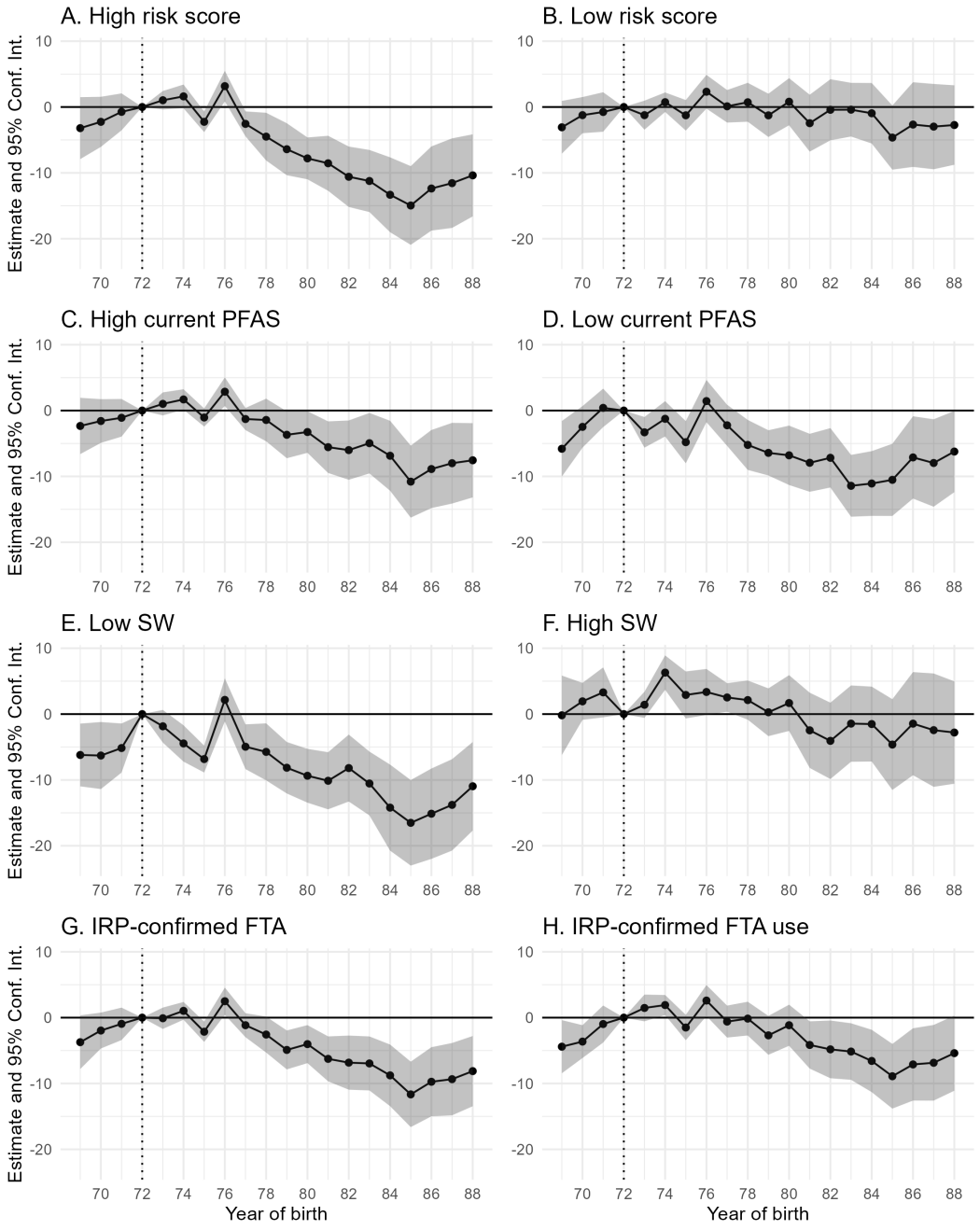


Figure A.4: Heterogeneity in exposure risk: birthweight

Note: This figure plots event study coefficients for cohorts born from 1969–1988, corresponding to Equation 1, for subsets of the data by proxies for exposure risk described in the main text. The dependent variable is individual birthweight in grams, the treatment variable is equal to 1 in FTA-exposed counties, and the baseline (before subsetting) estimating sample is individuals born in counties containing any military installation. In rows 1 and 2, the control group is unchanged across both specifications, and the "Low" designation includes counties or tracts for which information is not available. In row 3, both treated and control counties are divided by county surface water reliance. In row 4, treated counties without IRP-confirmed information are dropped. Regression controls include child and mother demographics, birth year-varying county weather and economic indicators, county fixed effects, and child sex-race-birth year fixed effects; see text for details.

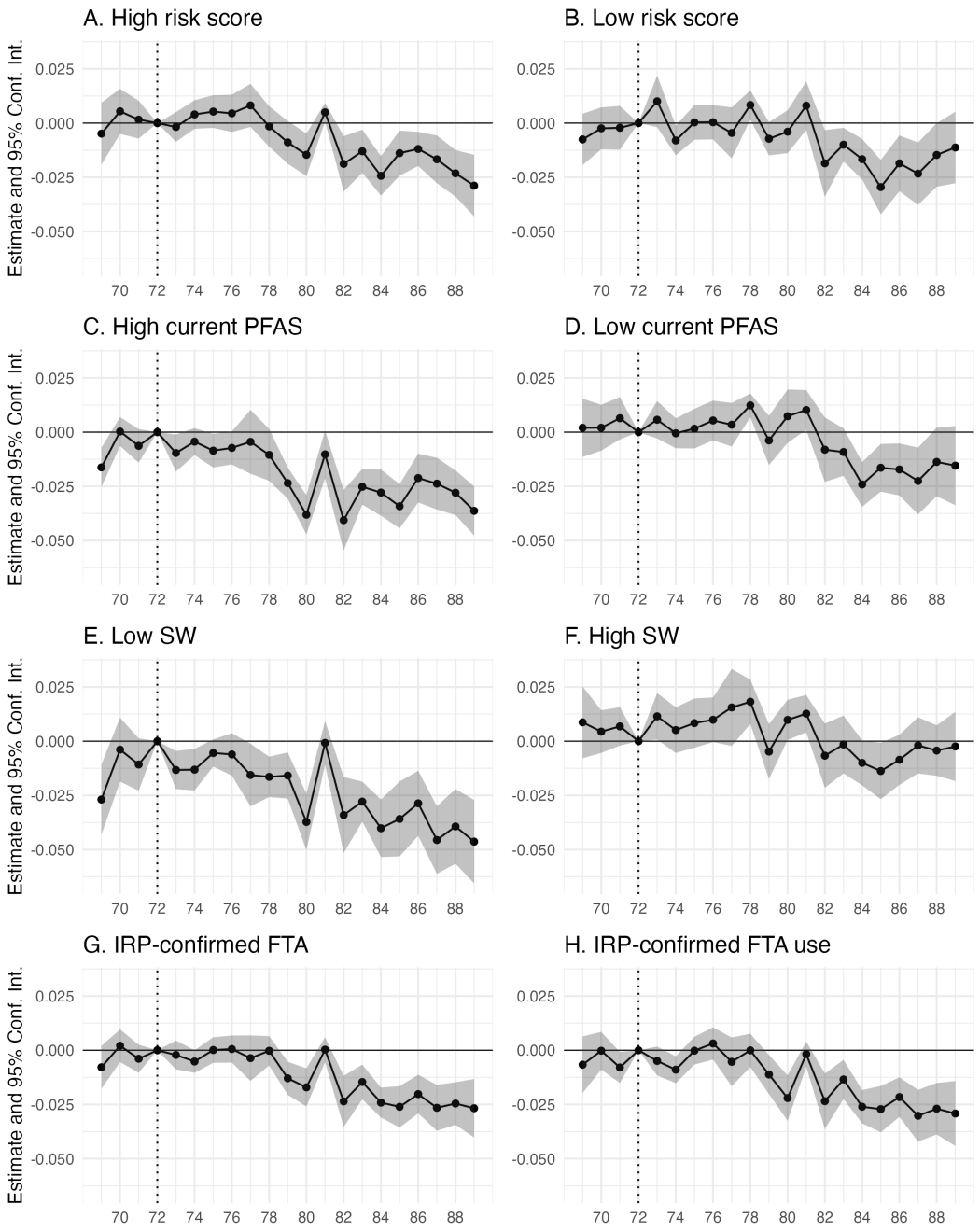


Figure A.5: Heterogeneity in exposure risk: earnings

Note: This figure plots event study coefficients for cohorts born from 1969–1989, corresponding to Equation 2, for subsets of the data by proxies for exposure risk described in the main text. The dependent variable is logged earnings, observed from 2015–2019, in 2019 USD, the treatment variable is equal to 1 in FTA-exposed tracts, and the baseline (before subsetting) estimating sample is individuals born within 30km of any military installation. In rows 1 and 2, the control group is unchanged across both specifications, and the "Low" designation includes counties or tracts for which information is not available. In row 3, both treated and control counties are divided by county surface water reliance. In row 4, treated counties without IRP-confirmed information are dropped. Regression controls include birth year-varying county weather and economic indicators, tract fixed effects, and individual sex-race-birth year-tax year fixed effects.

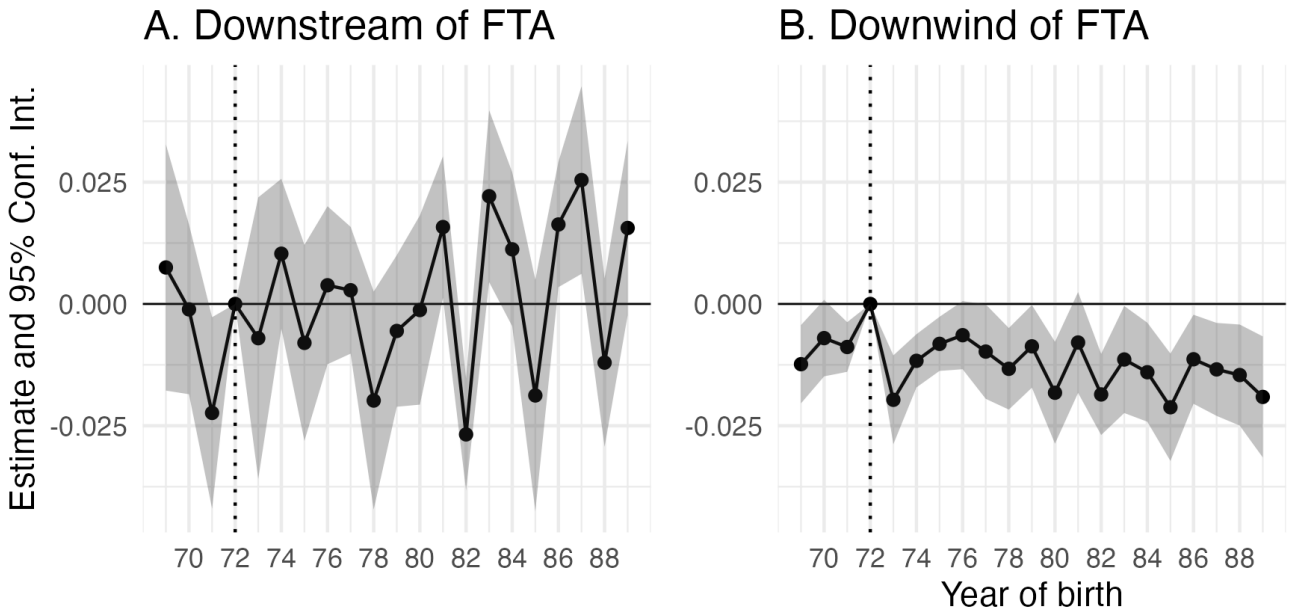


Figure A.6: Downstream and downwind FTA exposure

Note: This figure plots event studies for cohorts born from 1969-1989, subsets of the data by downstream (or downwind) tracts. The dependent variable is logged earnings. In Panel A, the treatment variable is equal to 1 in downstream-FTA tracts intersected by a downstream flowpath from an installation with an FTA. In Panel B, the treatment variable is equal to 1 in downwind-FTA tracts, corresponding to the cardinal direction of the prevailing wind direction from installation with an FTA. The estimating sample is restricted to tracts downstream (or downwind) of any installation. Regression controls include birth year-varying county weather and economic indicators, tract fixed effects, and individual sex-race-birth year-tax year fixed effects.

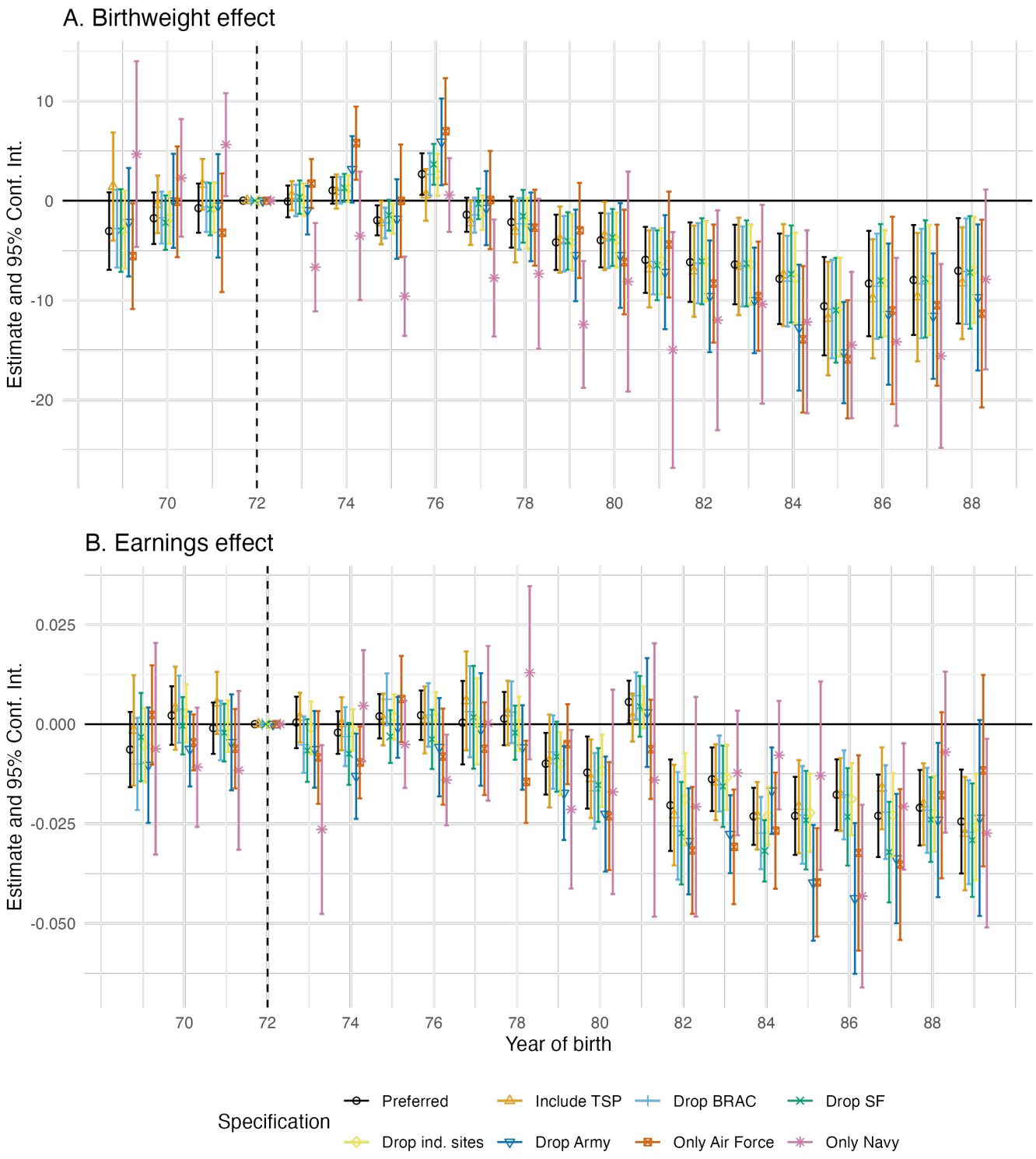


Figure A.7: Alternate sample subsets and controls

Note: Event study coefficients from sample subsets corresponding to possible threats to identification described in the main text. The dependent variables are birthweight for cohorts born in FTA-counties in Panel A ("Preferred" corresponds to Figure 1), and logged earnings for cohorts born in FTA-tracts in Panel B ("Preferred" corresponds to Figure 2B). The first alternate specification controls for county-level TSP. The second and third drop counties or tracts affected by BRAC closures and Superfund listings, respectively. The fourth drops counties containing the other major historic PFAS source of water contamination, industry sites. The remaining three specifications subset across different military branches, as described in Section 2.2.

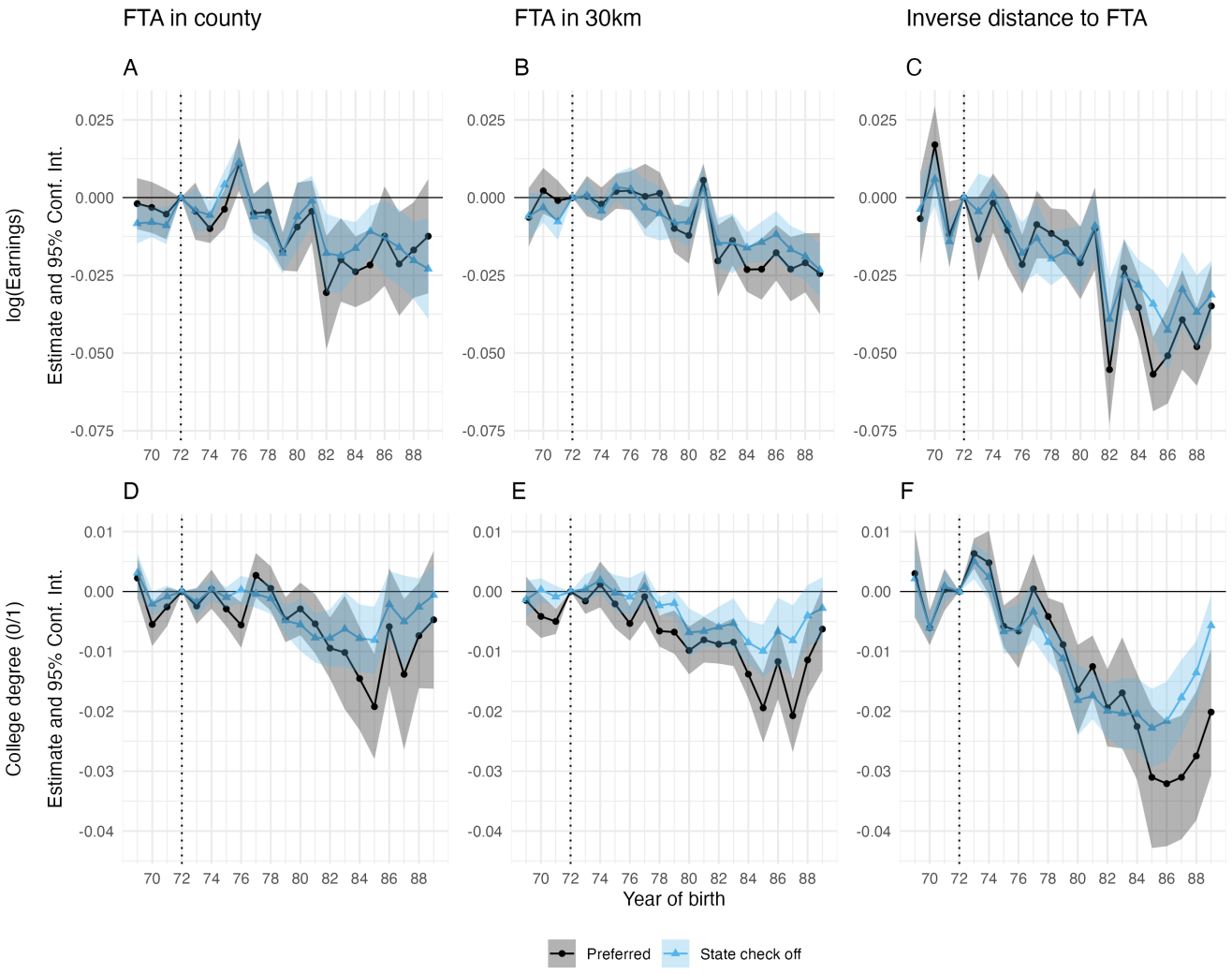


Figure A.8: Earnings and college completion effects, sample selection

Note: This figure modifies Figure 2 to show the effect of the preferred sample subset, which drops individuals with inconsistent state of birth data across the IRS records and Numindent survey. Event study coefficients for the estimating sample without this filter are shown in blue ("State check off").

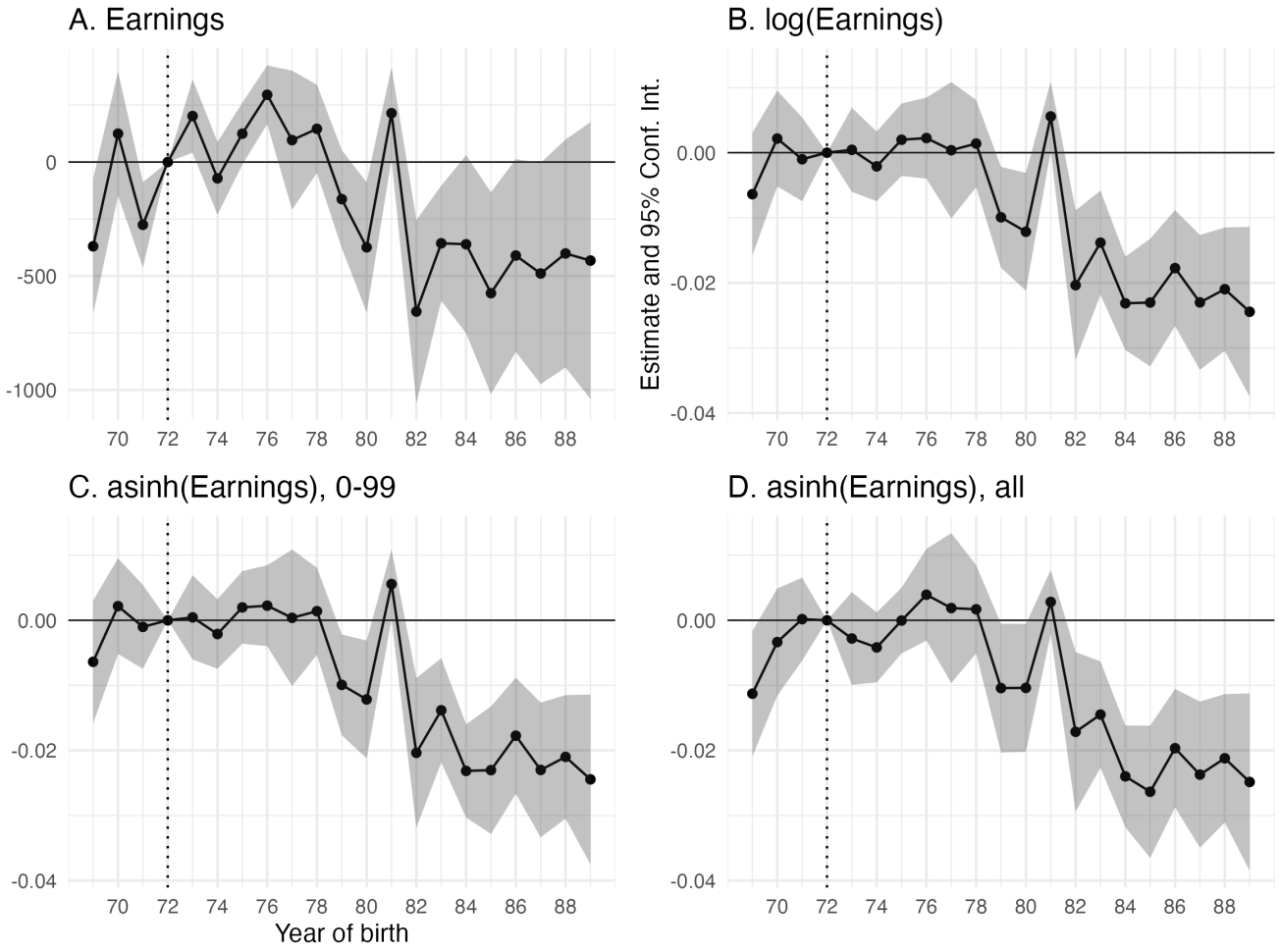


Figure A.9: Earnings effects, outcome trimming and transformation

Note: This figure shows how the event study in Figure 2B (here, Panel B) changes when using levels outcomes with the preferred sample (Panel A), the inverse hyperbolic sine transformation for the preferred sample (Panel C), or the inverse hyperbolic sine transformation without truncation (that is, including individuals with 0 reported earnings or earnings above the 99th percentile by age) (Panel D).

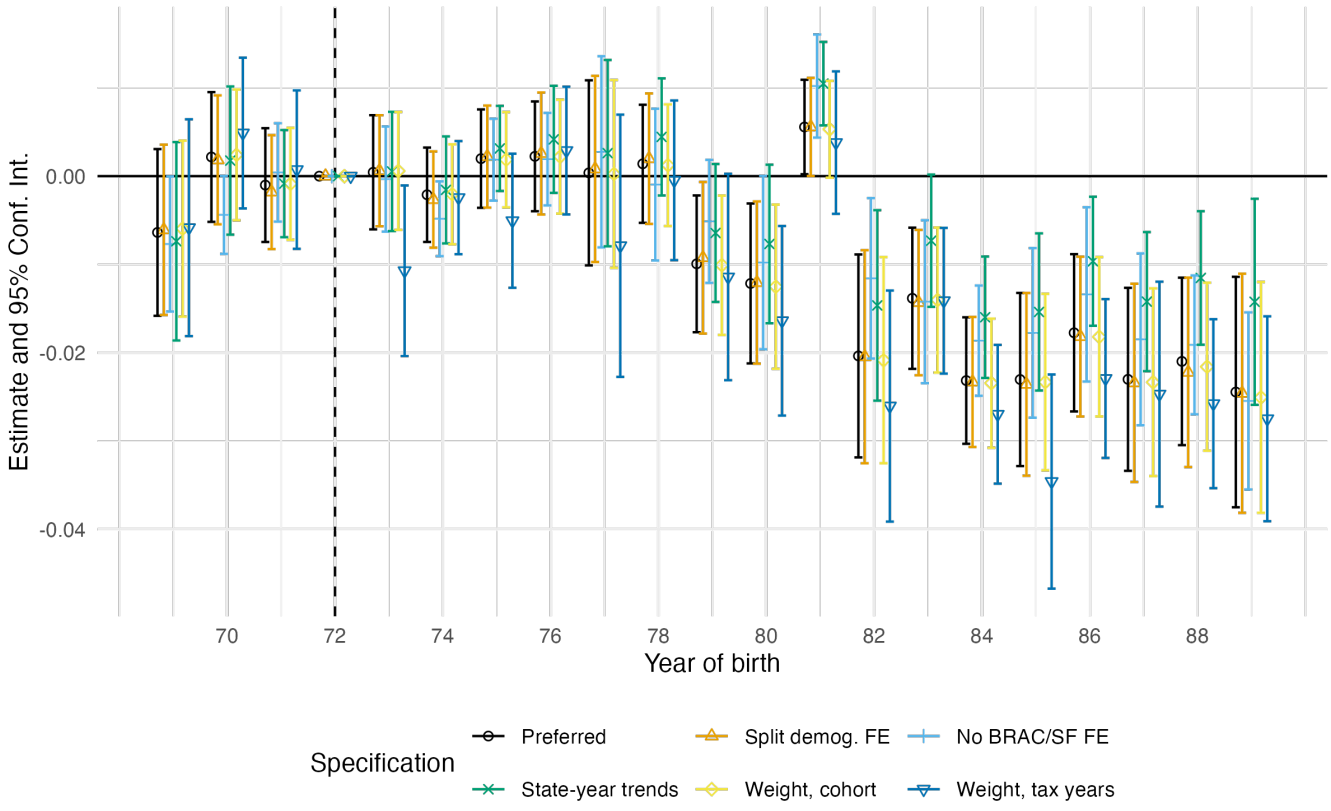


Figure A.10: Earnings effects, alternate controls and weights

Note: This figure shows how the event study in Figure 2B (here, "Preferred") changes for variations of the main specification. The first alternate specification splits the demographic FE from sex-race-birth year-tax year into separate sex-race and birth year-tax year FEs, and the second drops the set of FEs that interact child birth year with installation BRAC and Superfund listing dates. The third includes state of birth-year trends. The final two weight the specifications by inverse birth year cohort size (giving smaller cohorts higher weights) and inverse counts of individual years in the IRS earnings data (giving individuals with fewer earnings observations higher weights).

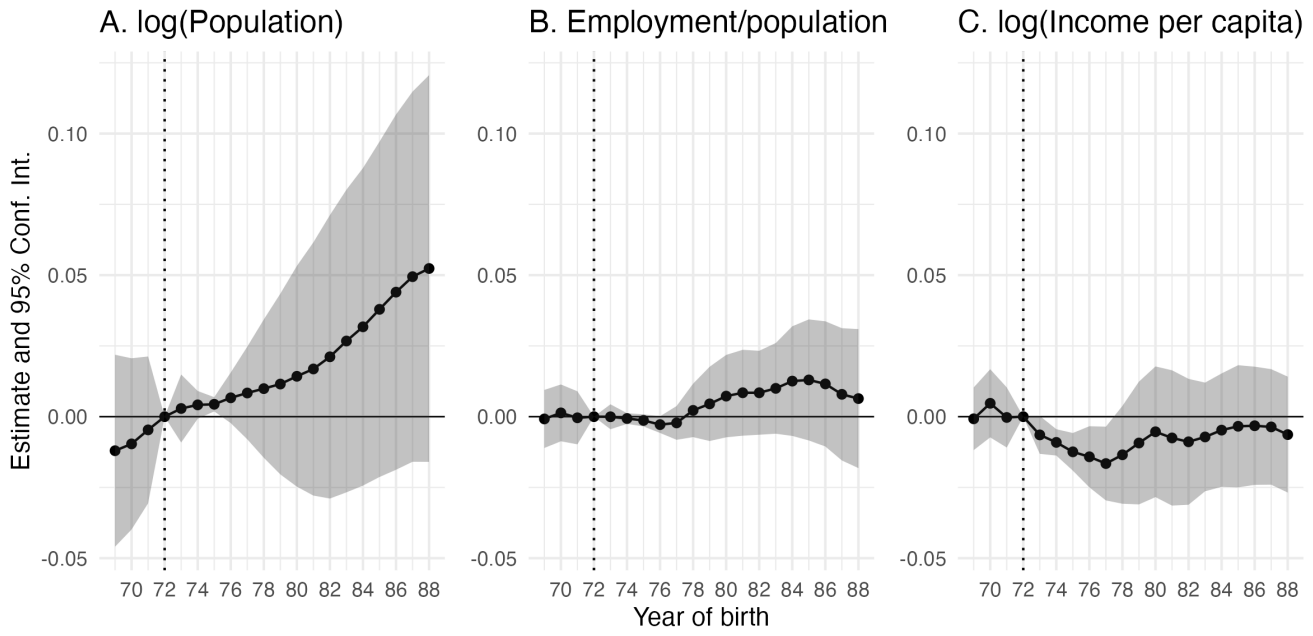


Figure A.11: County socioeconomic trends by FTA exposure

Note: This figure plots event study coefficients for county economic characteristics from 1969–1988. The dependent variable is logged county population in Panel A, the employment-to-population ratio in Panel B, and logged income per capita in Panel C. The treatment variable is equal to 1 in FTA-exposed counties, and the estimating sample is counties containing any military installation; observations of counties are weighted by their 1969–1971 population. Regression controls include county fixed effects and year fixed effects.

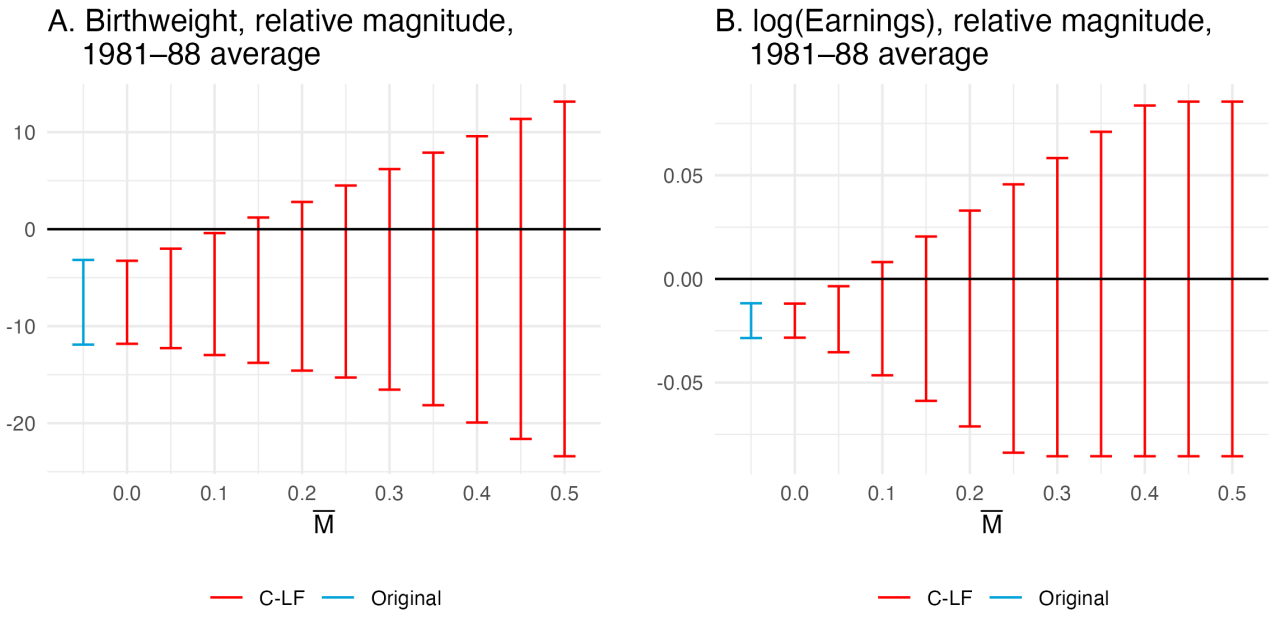


Figure A.12: Parallel trends sensitivity analysis

Note: Figures implement the Rambachan and Roth (2023) parallel trends sensitivity analysis, which bounds the post-period violation of parallel trends by $\bar{M}$ multiples of the maximum pre-treatment violation of parallel trends. Panel A uses the specification and sample described in the text and shown in Figure 1, and Panel B uses the specification and sample described in the text and shown in Figure 2B. The blue confidence interval ("Original") corresponds to the average effect for the later post-period (1981–1988) birth cohorts for each specification, assuming parallel trends. The red confidence intervals ("C-LF") show the effect of increasing $\bar{M}$, and indicate results are robust to post-period violations up to about 0.1 times the maximum pre-period violation of parallel trends.

B Data appendix

Our birthweight and earnings panels combine public and restricted data to assign PFAS exposure to individuals born in counties with military installatios across the United States.

B.1 Military installation data

B.1.1 Core installation data

We create a dataset of military installations using publicly available sources. To geolocate military installations from the DOD list (US DOD, 2021b), we hand-combine observations from the Military Installations with Ranges and Training Areas shapefile, a Census Bureau TIGER/Line shapefile, the Formerly Used Defense Sites shapefile, and the Bureau of Transportation Statistics's military shapefile dataset. A small number of installations on the 2021 DOD list did not appear in any geospatial data and for these, a latitude and longitude approximating the installation's historic location were hand-researched and entered.

We combine geolocated military data with the inventory of fire-crash training area sites (US DOD, 2014). When a county or tract is proximate to multiple installations, we consider the location to be exposed to fire training activity if one or more installations had an Air Force or Navy FTA. We include in our dataset of AFFF-exposed counties any county where greater than one percent of an installation's area falls within the county. Occasionally, multiple sites with distinct missions appear to be associated with the same geospatial polygon. We match FTA and other site-specific information (such as closure dates) before collapsing to unique polygons. There are 681 distinctly named sites associated with 640 unique polygons.

The Base Realignment and Closure (BRAC) process began in 1988 and accelerated in the 1990s; the final wave of closures was in 2005. The DOD BRAC Commission discloses the planned closures and projects in reports to the President, which are publicly available. We match our military sample with all five waves of BRAC documents to identify that experienced closures (Defense Base Closure and Realignment Commission, 1988, 1991, 1993, 1995, 2005). Closures affect about 20 percent of the sites in our sample. We construct a time-varying share of closures in a county to account for the economic impact of closures. The expected direction of BRAC effects on long-run economic outcomes is ambiguous. In the literature, Dardia et al. (1996) finds base closure effects are highly localized, and larger when installations are located in smaller communities. Krizan et al. (1998) finds "workers' employment prospects improve as bases shrink and the effect is actually stronger in smaller towns."

Under the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA, also known as Superfund), the EPA maintains a National Priorities List (NPL) identifying sites in the U.S. contaminated with hazardous substances and pollutants and slated for remediation. Federal sites are included in NPL, and we match listed sites and listing dates with our military data (US EPA, 2024c).

B.1.2 Installation Restoration Program

We use historical Installation Restoration Program reports to corroborate our FTA data. The first phase of the IRP inventoried risks in site investigation reports for active and formerly used installations. Site investigation reports were completed by environmental consulting firms in the 1980s, and provide selected descriptions of the existence, use, and characteristics of FTAs in the 1970s and early 1980s. Of environmental concern at the time was not the potential for groundwater contamination from AFFF, but the release of the jet fuel and other chemicals ignited for fire training, a practice that started before the switch to AFFF. Unfortunately, limitations exist: while many IRP reports from sites on our DOD list mention FTAs, most

are not specific about the frequency and intensity of training; further, some installations' Phase I reports are not publicly available (disproportionately Navy installations). Effluent management practices at FTAs are infrequently mentioned in the reports, but we find evidence of FTAs in use in the 1970s without berms or impervious liners, leading to uncontrolled runoff and infiltration. An example from the 1985 Grand Folks Air Force Base in North Dakota IRP: Phase I report: "Training activities at GFAFB include firefighter training. Fire training exercises are conducted regularly using jet fuel (JP-4) as fuel and using water and aqueous film forming foam (AFFF) as a suppressant." confirms the usage of AFFF and FTA for the base.

We speculate from our reading of the historic inspections that training frequency may not neatly correspond to contamination potential: the least frequently used FTAs were much less likely to have any effluent controls than high-use FTAs. It is also possible the reported training frequencies, which were recalled about a decade after (often by on-site fire chiefs), are inaccurate. They are infrequently available.

B.1.3 Recent Site Reports and Groundwater Sampling

The DOD has been investigating PFAS risk and initiating the clean-up process at installations. The DOD's PFAS task force was created to inform the public about the process and risks (Office of the Under Secretary of Defense for Acquisition and Sustainment, 2024). Each military branch maintains a website to shares information with the public about installation characteristics, possible contamination pathways, existing of groundwater wells, and surrounding populations. These site investigation reports provide some official information about the possibility of heterogeneity in exposure risk.

To quantify this in a standardized way, we create a rubric and hand-score DOD reports for installations managed by the Army (US DOD, 2024a), Navy (US DOD, 2024c), Air Force (US DOD, n.d.), and National Guard (US DOD, 2022). When tracts or counties are exposed to multiple sites with available risk scores (or groundwater concentrations), we add them to reflect cumulative risk.

DOD site report formats vary with the branch of the military, and detail varies across branches and sites. The reports typically identify whether nearby groundwater wells exist, who the wells serve (commercial, private domestic, or public), and whether the wells are hydrologically downstream of the installation. Reports exist for 507 sites in our sample. At the time of data collection in Summer 2023, 174 sites in our sample either did not have not publicly available reports or investigations were ongoing. Given that the primary mode of AFFF transport is through water, we focus on information about the presence of groundwater drinking wells and their quantity, connectivity to the public water system, the estimated size

Table B.1: Site FTA status by branch, BRAC, and Superfund status

	No FTA	Any FTA
Air Force	49	160
Army	243	86
Navy	61	74
Other	6	2
BRAC	42	77
Superfund	56	86
Distinct sites	359	322

of the affected population, and whether they are situated in the hydrologically downstream area of an FTA. Extracts from site reports are reproduced in Table B.2, and the rubric in Table B.3 shows how we process site reports to generate exposure risk scores.

Data on current groundwater samples collected by the DOD were publicized and standardized by the nonprofit Environmental Working Group from public DOD reports and Freedom of Information Act requests to the DOD (Environmental Working Group, 2024). The nonprofit publishes (in map form) a compilation of the maximum concentrations of PFAS detected in on-base groundwater between 2016 and our data collection time (summer of 2023) as a map. We hand collected and matched the sampling data from EWG's map to our list of installations. Testing tends to over-sample the most polluted sites, while sites determined to be less likely to be contaminated are scheduled for future testing. We have PFAS maximums for 296 bases in our sample.

Table B.2: DOD site report examples

Site name	Description
Fairchild AFB	"The former FTA was used after 1970 through approximately 1991. During this period, an estimated 125 gallons or more of AFFF were used during each training exercise. After each exercise, the remaining water, fuel, and foam were drained into an OWS, which discharged effluent water into a low area east of the training location. Based on the results of ongoing off-base sampling of private and public water supply wells and existing monitoring wells, PFAS in groundwater have migrated off-base into the adjacent community and into unincorporated areas. The current off-base area extends up to approximately 6 miles from the base (to the northeast) and includes 372 private wells and four municipal wells (API 2019a)" (US DOD, 2024d).
Boise Air Terminal	"The PA Report (C2HM 2015) indicates approximately 870 groundwater wells are located within a 3-mile radius of the former FTA (PRL 1), including private wells, Suez Water Idaho domestic water supply wells, and groundwater monitoring wells (EDR 2015). Suez Water Idaho owns and operates a domestic water well at Gowen Field (Byrd Well), which is approximately 2,700 ft upgradient from the former FTA. The Suez Water Idaho Byrd Well has not been used since 2005 (C2HM 2015). An additional off-Base well (Mac Well), owned and operated by Suez Water Idaho, is located approximately 4,000 ft west of the western boundary of Gowen Field (approximately 2 miles west and upgradient of the former FTA). The entire Suez Water Idaho system serves approximately 214,000 people, as well as Gowen Field (C2HM 2015)" (US DOD, 2024f).
Ellington Field	"The facility's drinking water is supplied by the City of Houston. A query of the Texas Water Development Board Submitted Driller's Reports and Groundwater Database identified four industrial wells, thirty-two (32) monitoring wells, four public supply wells, and one domestic well within a 1-mile radius of the facility. The industrial water supply wells range in depth from 376 feet to 555 feet bgs. The monitoring wells range in depth from 15 to 90 feet bgs. The public supply wells range in depth from 391 to 583 feet bgs, and the domestic well is 548 feet bgs. The public supply wells are owned by Time Warner Cable, BW Grayson Business Park, and the City of Houston Ellington Field (two wells). Humble Oil & Refining Co. own one domestic well. These wells are shown on Figure 1-2." (US DOD, 2024e).

Table B.3: Rubric for PFAS exposure potential

Points	Factor	Explanation
1	> 0 people affected	If the site reports identify any population of interest (either those who rely on affected water wells or residential populations in the focus area), we consider the affected population is greater than zero. If the report does not mention population but contains only water characteristics, this factor will be zero. If the report points to an ambiguous number of people who are more likely to be affected, we consider it greater than zero but less than 10,000.
1	$\geq$ 10k people affected	When the site report explicitly gives the number of people who are most likely to be impacted and this number is greater than 10,000
1	$\geq$ 50k people affected	When the site report explicitly gives the number of people who are most likely to be impacted and this number is greater than 50,000
1	Downstream wells exist	The literature states downstream concentrations are significantly higher than upstream, and it is less likely that PFAS reach hydrologically unconnected areas. Thus, if the site report indicates any downgradient well in the focus area, we consider downstream wells are present. If the well is in the focus area but upgradient or side gradient, we consider there are no downgradient wells in the respective base.
1	Public/municipal wells	If identified wells are part of public water systems, municipality water systems or utilities
1	> 0 wells affected	The potable well definition includes drinking wells (private or residential), domestic wells, potable water supply, or wells that are a part of the public water system. If all existing wells are used for monitoring, irrigation, commercial, or industrial purposes, we conclude that no potable well exists in the focus area. The site report provides information about the number of potable wells in the focus area. When the report is ambiguous about the number of wells (use language such as several, numerous, or multiple wells), we consider that the number of wells is greater than zero but less than 10.
1	$\geq$ 10 wells affected	When the site report explicitly states the total number of potable wells in the affected area, the number is greater or equal to 10.
1	$\geq$ 75 wells affected	When the site report explicitly states the total number of potable wells in the affected area, the number is greater or equal to 75.

B.2 All other data sources

B.2.1 Natality

Vital Statistics Natality Birth Data from 1969 to 1988 are publicly available at the county level for the U.S. (National Center for Health Statistics, 2014). We include child demographics and collapse the child race variable to three categories (white, Black, and all others), and we drop individuals for whom any of birthweight, child sex, child race, or mother's age are not observed (0.2 percent of the individuals born in counties with installations). Child birthweights are reported in grams, and gestational age is reported in weeks. We code indicators for low birthweight for observations under 2,500 grams and premature births for gestational lengths under 37 weeks. Gestational age is not reported for 18 percent of the observations in our sample that do report birthweight. Throughout our analysis, we omit individuals born in Alaska, Hawaii, or Puerto Rico.

B.2.2 Administrative Data

We link parents to children by combining dependent-child links from the 1994 tax returns with the probabilistic links from Stinson et al. (n.d.), which gives us coverage of birth cohorts from the 1960s through the mid-1990s. Because historic tax years do not cover every birth year in our study period, we expect but cannot quantify measurement error in tract of birth due to parental moves. The tax returns are geocoded to the home address level, which we aggregate to Census tract boundaries, and contain the SSNs of dependent children. We omit from the estimating sample

We assign wages at the individual level, and AGI at the tax unit level (capturing family income for married couples, and individual income for non-married individuals). All income is inflation-adjusted to 2019 dollars. College attainment was coded to indicate individuals with a completed four-year degree. We drop individuals with inconsistent state of birth data across the IRS records and Numindent survey; these may differ due to household mobility in very early childhood or administrative errors. In earnings regressions, we drop individual-tax year observations where an individual did not receive a W-2, received a W-2 with annual earnings equal to 0, or had earnings that exceed the 99th percentile by age.

B.2.3 U.S. Geological Survey

We construct the downstream surface flows for each base in our sample using the National Hydrography Dataset. Because we do not know the exact location of FTAs within sites, we construct a grid of 100 points within each installation's boundary as origin points to trace downstream paths. We identify downstream tracts as those intersected by a downstream flowpath from any nearby installation, and an FTA-exposed downstream tract as those intersected by a downstream flowpath from an installation with an FTA.

We obtain county-level drinking water shares from the USGS (USGS, 1985). We construct the share of the county population served by surface-sourced public water as the ratio of the reported population in that group and the county total.

B.2.4 Local meteorology

We obtain county-level minimum and maximum temperature and precipitation for our study period from Schlenker and Roberts (2009). We aggregate these to county-year mean temperature and precipitation.

We use wind data to identify tracts in the prevailing downwind direction from installations. The North American Regional Reanalysis data set for 1979–1990 provides monthly mean values

for east-west and north-south components of wind for a 32-by-32 kilometer grid. We define downwind tracts as those in the 90-degree cardinal direction bin of the prevailing wind direction in more than 50 percent of year-months for any site. We convert u-v components to a 0–360 scale and bin these to 4 cardinal coordinates (north, east, south, and west). We then calculate the relative frequencies of each cardinal direction bin for the study period for each site. When a military site intersects more than one grid, the weighted average of grid values in which centroids fall inside to location boundary is taken.

B.2.5 Economic indicators

We use time-varying county-level economic indicators from Bureau of Economic Analysis. Our analysis include county population, per capita personal income, and employment to population ratio from the CAINC4 "Personal income and employment by major component" table (US Bureau of Economic Analysis, 2023). We include income per capita, population, employment per population per county for each year in our study period.

Non-military sources of PFAS

At military installations without FTAs, AFFF would have been installed in fire suppression systems in areas like aircraft hangars, kept in crash response vehicles, and used in cases of actual crashes. Non-military sources of PFAS included industries that produce and handle the chemicals, wastewater treatment plants, incinerators, and landfills; each of these contribute to air, water, and soil contamination. Water exposures represent a meaningful share of exposures from all pathways,[29] and FTAs are often cited as the dominant source of legacy water contamination (Sunderland et al., 2019; Banzhaf et al., 2017; Dasu et al., 2022). Hu et al. (2016) finds relatively small PFAS contamination effects from wastewater treatment plants, and non-significant effects from civilian airports, but significant associations between military FTAs and PFAS contamination. That study also identifies another major, early source: 16 manufacturing sites listed in the U.S. EPA's 2010/2015 PFOA Stewardship Program. These sites had been chosen for major PFAS manufacturing and possible contamination. We collect the counties in which these manufacturing sites were located to test robustness of main results to their exclusion.

B.2.6 Pollution data

The EPA's Toxics Release Inventory began collecting data from major sources of industrial chemical pollution in 1987. Because the 1988 reporting year is substantially more complete, we use this year's reports to characterize differences in exposure to toxic releases across treatment and control counties at the end of our study period, noting that the mix of facilities operating in the 1970s may have been different. Counties exposed to FTAs had, on average, more TRI facilities than non-exposed counties and higher total on-site releases, but toxicity-weighted releases were several orders of magnitude higher in control counties. We aggregate on-site releases from reporting facilities to county-level pounds of toxic releases, and we additionally construct toxicity-weighted total releases using the EPA's toxicity weights for each chemical (taking the maximum of the oral or inhalation toxicity weight). FTA counties reported 1,137,143 total pounds of reported releases in 1988, and non-FTA counties reported 891,038. Toxicity-weighted total releases were several orders of magnitude higher in control counties.

We query EPA's Air Quality System API for total suspended particulates by county and year, averaging across all reporting monitors available for a given year. Many counties, partic-

[29] The EPA estimates relative source contributions (RSC) to characterize exposure pathways, using a 20 percent RSC for drinking water in 2016; several states go higher: New York uses 60, and Michigan, New Hampshire, Minnesota, and Washington use 50.

ularly those in rural locations, are not monitored by the EPA and so are missing these data. We use the EPA's Green Book data for yearly county nonattainment status.

Keiser et al. (2023) provides the number of grants under the Clean Water Act received by facilities. We sum across facilities located in study counties to find the total number of CWA grants issued per county.

C Epidemiological evidence for health effects of PFAS

A growing medical literature studies the link between PFAS, especially PFOS and PFOA, and adverse health outcomes. This work has focused on both fetal and post-natal growth restrictions, but also explored adverse health outcomes in adults. Many of the studies are associational in nature, but as a group they suggest serious and potentially lasting effects on human health from exposure.

Studies of the effect of exposure to PFAS compounds on fetal growth restriction (such as diminished birth weight, gestational length, size for gestational age, and head circumference) have had mixed results. Most find evidence of reduced birth weight from exposure (Ashley-Martin et al., 2017; Xu et al., 2019; Blake and Fenton, 2020), while some find null (Bell et al., 2018) or imprecise (Meng et al., 2018) outcomes. Sagiv et al. (2018) finds an impact on gestational length, and Hjermitslev et al. (2020) finds an adverse effect on birth weight-length and head circumference.

Early childhood exposure to PFAS may also affect post-natal growth. There is evidence of an association between immunosuppression and asthma incidence in children (Jackson-Browne et al., 2020; Granum et al., 2013). Hoffman et al. (2010) finds a positive association between ADHD-like behaviors and decreased executive function in children in highly exposed communities, while the findings in Stein et al. (2013) are inconclusive. The EPA has described the existing literature insufficiently conclusive (US EPA, 2016b,a).

Less is understood about the health effects of PFAS exposure on the adult population. Diabetes, cardiovascular illnesses, and serum lipids problems have been studied without producing definitive evidence (US EPA, 2016b,a). However, a recent epidemiological literature review Bartell and Vieira (2021) suggests an association between exposure and both kidney and testicular cancer. In a toxicological review of exposure to PFAS, Blake and Fenton (2020) points to adverse effects at birth and from occupational exposures, and suggests these may lead to long-run susceptibility to later health conditions, but suggests much more research is needed to understand these long-run effects.